No Country for Honest Men: Balanced Theft, Endogenous Patronage, and the Emergence of Enforcement

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Abstract

A unit transfer raiding game can sustain equality when each household leaves home, each unoccupied house is targeted once, and concurrent raiders would otherwise divide the same loot. *The Black Sheep* turns on the breakdown of that reciprocal order, and this paper formalizes the sequence. In the benchmark stage game, a balanced theft cycle is a weak Nash equilibrium. In the repeated benchmark with stationary Markov behavior, one permanently honest household stays home, blocks the raid directed at it, and creates a first-night windfall for the adjacent successor in the old cycle. If leaving home exposes current wealth, that windfall moves the successor above the threshold at which personal raiding ceases to be optimal, so the original cycle cannot persist as a stationary continuation. A proportional loot extension then studies the later transition. Rich households hire thieves when delegated predation yields surplus over the worker outside option and contracting cost, and they later replace thieves with guards when concentrated wealth and prey depletion raise the value of protection above the value of one more round of predation. Simulations implement the same payoff comparisons and record the timing of class differentiation, patronage, selective protection, inequality, and welfare loss once theft, contracting, guarding, and punishment consume resources. A local honest rupture can therefore dissolve a reciprocally corrupt order without producing an impartial institutional exit.

Keywords: repeated games, corruption equilibria, patronage, endogenous enforcement, inequality

JEL classification: C73, D74, K42, O17

1 Introduction

In *The Black Sheep*, Calvino begins from a society in which theft is universal and equality survives only because every household is both predator and prey. Losses and gains offset one another exactly,

so the social order remains predatory without becoming stratified. The story changes when one man stays at home. The raid directed at his house fails, the house he would have robbed remains untouched, and a local disturbance in the transfer network appears at once. The later movement toward hierarchy, delegation, and policing follows from that break (Calvino, 1993). The opening order can be written as a unit transfer raiding game in which each household chooses in every period whether to stay home or which other house to raid, occupied houses are safe in the benchmark, and concurrent raiders divide a rival loot whose expected share falls with congestion. Under a balanced theft cycle every household leaves home, every unoccupied house is targeted once, and wealth remains level because each household has one incoming loss and one outgoing gain. Universal predation is then compatible with equality through exact accounting, and that opening order survives as a weak Nash equilibrium of the stage game.

In the repeated version, the horizon is infinite, behavior is restricted to stationary Markov strategies, and one household is permanently honest and therefore remains at home in every period. If the remaining households follow the old cycle on the first perturbed date, the predecessor who targets the honest house returns empty handed, the successor escapes the raid that would otherwise have hit him, and the resulting wealth change is local but exact. Once leaving home also exposes current wealth, the successor's windfall can move him above the cutoff at which personal raiding ceases to be privately attractive, so the original balanced cycle no longer survives as a stationary continuation. A richer environment becomes necessary at that point, because exact unit transfers cannot carry hierarchy very far once wealth begins to diverge. The extension therefore preserves the same population, the same public wealth state, and the same logic of explicit target choice while replacing unit loot with proportional loot and enlarging the occupational menu. Rich households compare personal raiding, staying home, hiring thieves, and later hiring guards; poor households compare freelance theft, paid predation, protection work, and withdrawal. Patronage requires delegated predation to reach targets more effectively than freelancers can and poor workers to face outside options weak enough to leave a surplus after wages and contracting cost. Guarding appears later, once concentrated wealth and prey depletion alter the ranking of control technologies. Hierarchy, delegation, and selective protection then emerge as

successive environments, each tied to its own payoff comparison and transition condition.

Several literatures meet naturally at that point. Corruption equilibria remain privately coherent because each participant takes as given an order that depends on the participation of others (Andvig and Moene, 1990; Bardhan, 1997; Shleifer and Vishny, 1993; Tirole, 1996). Local perturbations then matter through the same logic that convention and equilibrium selection models use to study movement across sustained orders (Young, 1993). Appropriation and defense enter once predation and protection compete for resources under insecure claims (Skaperdas, 1992; Grossman and Kim, 1995; Hirshleifer, 1995). Hierarchy enters once the rich use vertical organization to redirect predatory labor (Olsen and Torsvik, 1998). Organized enforcement enters once protection stabilizes an unequal order created during the transition itself (Olson, 1993; North et al., 2009; Besley and Persson, 2009; Acemoglu et al., 2011). The graph of the balanced benchmark also recalls a directed exchange cycle, which is why Shapley and Scarf (1974) remains useful as a limited structural reference. The connection stops at the graph. Transfers here are coercive, occupancy changes the opportunity set of others, and rivalry in extraction creates externalities that do not belong to the voluntary exchange environment.

Aggregate wealth cannot be the welfare object because the benchmark preserves it by construction. Period welfare is therefore defined as the sum of concave utilities over wealth minus theft effort costs, contracting costs, guarding costs, and expected punishment losses. That criterion distinguishes pure transfer, inequality, and resource dissipation, and it also keeps the simulations in their proper place. No full Markov perfect equilibrium is solved for the large stochastic state space generated by matching, delegated predation, and protection,[1] and production remains outside the model. The simulations accordingly implement the same action logic, targeting rules, and threshold comparisons used in the theory, then measure transition timing, regime prevalence, inequality, and welfare along paths that the analytical results do not characterize in closed form. Section 2 places the argument in the relevant literatures. Section 3 presents the unit transfer game, the repeated perturbation result, the proportional extension, the contracting problem, the guarding threshold, and the welfare criterion. Section 4 states how those analytical objects are mapped into code. Section 5 presents the figures and tables in the

[1] The theory isolates the margins that can be derived cleanly. The simulation section studies how those margins interact once the extended state space becomes too large for complete analytical characterization.

order required by the argument. Section 6 closes.

2 Related Literature

The setting in Calvino belongs to a familiar concern in economics, namely the possibility that conduct which is plainly undesirable in welfare terms can nevertheless form a coherent and durable equilibrium. In the corruption literature this concern appears in several adjacent forms. Andvig and Moene (1990) show how corruption reproduces itself by altering the incentives faced by later actors. Bardhan (1997) emphasizes the institutional complementarities that make corrupt systems persistent. Shleifer and Vishny (1993) show that the social effects of corruption depend on whether predation is fragmented or coordinated, while Tirole (1996) studies how collective reputations trap organizations in persistent patterns of misconduct even when many participants would prefer a cleaner order. The same concern appears here in a narrower and more sequential setting. The object is a world of reciprocal thieves and the transition set off by a small honest rupture that disturbs the accounting on which reciprocity depended.

The same disturbance also places the analysis near the literature on conventions and equilibrium selection. Young (1993) studies how recurrent interaction and local perturbations move a society from one sustained convention to another. The central insight is that a local change can shift the continuation incentives of neighboring actors and thereby alter the basin of attraction of the entire system. The opening equilibrium in Calvino has exactly that structure. Its persistence is behavioral rather than normative. Under symmetry it is regular and reproducible even though it rests on universal predation. The first honest agent interrupts a pattern of mutual dependence, and the first windfall together with the first shortfall changes who still finds participation privately worthwhile.

The movement from reciprocal predation to hierarchical predation requires a different body of work. Much of the corruption literature begins with an organization that is already stratified and then studies collusion, monitoring, or internal incentive design. Olsen and Torsvik (1998) are especially useful because hierarchy there alters both the distribution of rents and the feasibility of corruption itself.

The same problem arises once rich households stop stealing personally and begin to buy predatory labor from poor households. The difference is that hierarchy is not assumed at the start. It has to be generated from the breakdown of a decentralized arrangement. Patronage therefore requires a strict surplus condition. Delegation cannot rest on vague appeals to elite power alone. It must be profitable because a patron directs theft more effectively than a freelancer can and because the reservation utility of poor workers remains compressed.

Skaperdas (1992) and Grossman and Kim (1995) study environments in which insecure claims induce agents to allocate resources between conflict and defense. Hirshleifer (1995) sharpens the same theme by analyzing the conditions under which anarchy unravels and more selective forms of order emerge. Demsetz (1967) remains a useful background reference because property rights there arise when internalization becomes valuable enough to justify protection. The connection is close in its treatment of appropriation and protection as alternative uses of resources, but the point of departure matters. The starting point is an order in which insecurity of claims is universal, reciprocal, and roughly egalitarian. The later rise of guarding is therefore not the first appearance of order. One order replaces another, and the later order protects a distribution created during the transition itself.

The connection to state formation and endogenous enforcement is equally direct. Olson (1993) distinguishes roving bandits from stationary ones and argues that an encompassing interest in future extraction can create incentives to provide order. North et al. (2009) generalize that intuition into a theory of violence orders organized around elite management of rents. Besley and Persson (2009) study the conditions under which legal and fiscal capacity emerge, while Acemoglu et al. (2011) show that inefficient coercive states can persist once they serve the interests of those who control them. Acemoglu and Robinson (2008) likewise stress that institutional change often preserves elite power. The final regime belongs in that lineage. Guards and police arise because concentrated wealth becomes worth defending and because labor can be reallocated from predation to protection without changing who controls the terms of social order.

Once guarding enters the picture, Becker (1968) and Becker and Stigler (1974) also become relevant because elite households no longer face only the choice between stealing and abstaining. No

benevolent state chooses optimal sanctions here, and no optimal policing problem is solved, but the comparison does shift toward two costly technologies, one extractive and one protective, each with its own labor demand and effect on future incentives.

Directed cycles matter here only as a graph theoretic analogy. A balanced theft cycle is also a directed cycle with one incoming and one outgoing link for each agent, and in Shapley and Scarf (1974) directed cycles organize voluntary exchange among agents who reassign indivisible objects. The analogy is useful at that level of structure and no further. The transfers here are coercive, alter later incentives through exposure and depletion, and impose externalities on actors who have not consented to any reassignment, so claims about the core, strategy proofness, or voluntary stability do not survive the change in environment.

Skaperdas (1992) and Grossman and Kim (1995) keep attention on appropriation and defense under insecure claims. Becker (1968) and Becker and Stigler (1974) keep attention on sanction, precaution, and the incentives of enforcement. Olson (1993), North et al. (2009), Besley and Persson (2009), and Acemoglu et al. (2011) show how organized coercion stabilizes extraction and protects concentrated advantage. Their intersection is the natural place for a setting that begins from egalitarian reciprocal corruption, because aggregate wealth cannot serve as the relevant welfare object once the benchmark preserves total wealth by construction. The comparison has to run through concavity in wealth together with the costs of theft, punishment, contracting, and guarding. Reciprocal corruption, hierarchical predation, and selective protection then appear as successive forms of one order, linked by a local honest rupture that shifts incentives without generating an impartial institutional exit.

3 Model

3.1 The unit transfer raiding game

The analysis begins from a repeated raiding game whose only purpose is to make the opening reciprocal order exact. Time is discrete and indexed by $t = 0, 1, 2, \ldots$. Agents are indexed by $i \in \{1, \ldots, N\}$ with $N \geq 3$, and wealth at the beginning of period t is the publicly observed vector

Table 1: Notation. The table collects the state variables, contract terms, and regime objects used throughout the theory and simulation sections.

Symbol	Meaning
$i \in \{1, \ldots, N\}$	Agent index.
$t = 0, 1, 2, \ldots$	Time index.
w_{it}	Wealth of agent i at the beginning of period t.
σ	Baseline theft permutation defining the reciprocal cycle.
h	Exogenously honest agent in the perturbation analysis.
τ	Unit loot in the benchmark stage game, normalized to $\tau = 1$.
γ	Proportional theft intensity in the dynamic extension, with $\gamma \in (0, 1)$.
c_T	Resource cost of personal theft effort.
ρ	Exposure coefficient capturing the expected fraction of own wealth lost when one leaves home.
ϕ_t	Targeting advantage of delegated theft relative to freelance theft.
s_{ijt}, α_{ijt}	Fixed payment and loot share in a patron-thief contract between patron i and thief j.
o_{jt}	Outside option of poor agent j if he declines a patron's contract.
k	Contracting cost borne by the patron.
g	Guard wage.
q_t	Probability that guarding blocks a raid on protected wealth.
R_t, E_t, P_t	Sets of rich, elite, and poor agents at date t.

$w_t = (w_{1t}, \ldots, w_{Nt})$. In the benchmark all agents begin with common wealth $\bar{w} > \tau$, where $\tau > 0$ is the unit amount that can be stolen from an unoccupied house in one period. The richer proportional environment appears only after the benchmark and keeps the same state variable. The notation used throughout the paper is collected in Table 1; the same symbols are used in the theory, the simulations, and the appendix proofs.

In each benchmark period, agent i chooses one action from the set

$$A_i = \{H\} \cup (\{1, \ldots, N\} \setminus \{i\}),$$

where H means staying home and $j \neq i$ means leaving home and attempting to raid household j. Occupancy determines vulnerability. If $a_{jt} = H$, household j is occupied and no raid on j succeeds in period t. If $a_{jt} \neq H$, household j is unoccupied and can be raided. This is the only occupancy rule needed for the benchmark, and it makes the first honest night deterministic once one household stays

home permanently.

Target congestion is explicit. For any household j, let

$$M_{jt} = \{i \neq j : a_{it} = j\}, \qquad m_{jt} = |M_{jt}|.$$

If $a_{jt} = H$, every attempted raid on j fails. If $a_{jt} \neq H$ and $m_{jt} \geq 1$, at most one unit τ can be extracted from j during period t, and that unit is divided in expected value equally across the raiders in M_{jt}. The benchmark therefore treats loot as rival and rules out the possibility that several agents can each take the same unit independently.

Define the realized loss of household i and the expected loot of agent i by

$$\ell_{it}(a_t) = \tau \mathbf{1}\{a_{it} \neq H\}\mathbf{1}\{m_{it} \geq 1\}, \qquad g_{it}(a_t) = \sum_{j \neq i} \mathbf{1}\{a_{it} = j\}\mathbf{1}\{a_{jt} \neq H\}\frac{\tau}{m_{jt}}. \tag{3.1}$$

The benchmark abstracts from direct theft cost so that the reciprocal accounting is exact. End-of-period wealth is therefore

$$w_{i,t+1} = w_{it} - \ell_{it}(a_t) + g_{it}(a_t). \tag{3.2}$$

3.2 Balanced reciprocity

Definition 3.1 (Balanced theft cycle)**.** *A balanced theft cycle is a permutation σ of $\{1, \ldots, N\}$ with no fixed points such that each agent i chooses $a_i = \sigma(i)$.*

Under a balanced theft cycle every household is unoccupied, every household is targeted by exactly one raider, and no target congestion appears because $m_j = 1$ for every j. The basic accounting is then immediate but worth stating explicitly because the rest of the paper measures departures from this object.

Lemma 3.2 (Balanced accounting)**.** *If all agents begin a period with common wealth $\bar{w}$ and play a balanced theft cycle, then every agent ends the period with wealth $\bar{w}$ and aggregate wealth is preserved.*

Proof. See Appendix A.1. Each agent loses exactly τ and gains exactly τ, so wealth is unchanged agent by agent and therefore in aggregate. □

Proposition 3.3 (Balanced reciprocity as a weak equilibrium)**.** *In the one-shot unit transfer stage game, any balanced theft cycle is a weak Nash equilibrium.*

Proof. See Appendix A.1. Deviating to staying home preserves the deviator's wealth but does not raise it, because the avoided loss is exactly offset by forgone loot. Deviating to a different target creates congestion, so expected loot falls below τ while the incoming loss remains. Hence no unilateral deviation is strictly profitable. □

Only a weak claim is needed at this stage. The purpose of the benchmark is not to establish efficiency or to characterize the full set of equilibria. It is to show that a reciprocal predatory order can be a coherent strategic object once actions and congestion are written down exactly. The resulting graph is a directed cycle, and that graph theoretic resemblance to top trading cycles remains useful when one link later breaks. The resemblance goes no further. Transfers here are coercive, occupancy changes the opportunity set of others, and rivalry in extraction creates externalities that have no analogue in the voluntary exchange environment of Shapley and Scarf (1974).

3.3 A permanent honest household and stationary continuation

The repeated benchmark keeps the same stage game and adds two restrictions immediately. The horizon is infinite with common discount factor $\delta \in (0, 1)$, and strategies are restricted to be stationary and Markov in current wealth. A strategy for agent i is therefore a map from the current state w_t into an action in A_i. This is the equilibrium class studied in the theorem below. The argument does not characterize the full set of repeated outcomes that could be sustained by history-dependent punishments.

Fix one agent h who is permanently honest and therefore chooses H in every period. Let σ denote the pre-perturbation balanced theft cycle, let $p = \sigma^{-1}(h)$ be the predecessor who planned to rob h, and let $q = \sigma(h)$ be the successor whom h would have robbed under universal participation. If every other agent continues the old cycle on the first perturbed date, then the occupancy rule makes the first

wealth change deterministic:

$$w_{h1} = \bar{w}, \qquad w_{p1} = \bar{w} - \tau, \qquad w_{q1} = \bar{w} + \tau, \tag{3.3}$$

and all other agents remain at $\bar{w}$.

Lemma 3.4 (First night local asymmetry)**.** *Under the benchmark cycle with one permanently honest agent h, the first honest night creates a local three agent asymmetry given by* (3.3)*. In particular, the successor q receives a windfall and the predecessor p suffers an immediate shortfall.*

Proof. See Appendix A.1. The honest agent neither steals nor is robbed, the predecessor fails in his attempted raid on an occupied house, and the successor is not robbed by the honest agent. □

The benchmark stage game is enough to show that exact reciprocity breaks. A withdrawal result needs one additional margin. Suppose that personal raiding in the repeated game carries direct effort cost $c_T \geq 0$ and expected exposure loss ρw_{it}, with $\rho \in (0, 1)$, because the raider leaves current wealth unattended while away from home. The expected one-period return from personal theft is then

$$\pi^S(w_{it}) = \tau - \rho w_{it} - c_T, \tag{3.4}$$

while the normalized return from staying home is $\pi^N(w_{it}) = 0$. Hence there is a wealth threshold

$$\widehat{w} = \frac{\tau - c_T}{\rho} \tag{3.5}$$

such that an agent weakly prefers personal theft when $w_{it} \leq \widehat{w}$ and strictly prefers staying home when $w_{it} > \widehat{w}$.

Assumption 3.5 (Interior exposure threshold)**.** *The exposure threshold satisfies $\bar{w} < \widehat{w} < \bar{w} + \tau$.*

The assumption places the symmetric benchmark wealth level below the exposure cutoff and the first-night windfall above it. Personal theft remains privately viable at $\bar{w}$ but ceases to be optimal after

one unit of windfall. This is a narrow comparative static statement and not a general theory of criminal choice.

Theorem 3.6 (No stationary one-honest continuation)**.** *Under Assumption 3.5, there is no stationary Markov equilibrium of the repeated benchmark game that preserves the original balanced theft pattern for all agents other than the permanently honest agent h.*

Proof. See Appendix A.1. By Lemma 3.4, the successor q enters period $t = 1$ with wealth $w_{q1} = \bar{w} + \tau$. Assumption 3.5 implies $w_{q1} > \widehat{w}$, so (3.4) yields $\pi^S(w_{q1}) < 0 = \pi^N(w_{q1})$. Thus q strictly prefers staying home at $t = 1$ and no longer continues to steal. Since the original balanced pattern requires q to keep stealing, that pattern cannot be stationary after the perturbation. □

The theorem is deliberately local. It does not solve the full repeated game with one honest agent. It says only that the original balanced cycle cannot survive as a stationary Markov continuation once the first honest night moves the successor across the exposure threshold. That is exactly the comparison later implemented in the deterministic benchmark simulation.

Chronology matters. The honest agent is not robbed immediately upon becoming honest because occupied houses are perfectly safe in the unit transfer benchmark. Only later, after the paper moves to the proportional environment with delegated predation and guarding, do occupied but unguarded houses become imperfectly safe. The deterministic first-night result and the later stochastic vulnerability therefore belong to different layers of the model and should not be conflated.

3.4 Proportional transfer extension and endogenous patronage

After the first honest disturbance, wealth differences matter directly for choice because leaving home now exposes different stocks of wealth. The extension therefore keeps the same population, the same public wealth state, and the same logic of explicit target choice, while replacing the unit transfer rule with proportional appropriation. If household j is successfully raided in period t, gross loot equals $\Gamma_{jt} = \gamma_j w_{jt}$, where $\gamma_j = \gamma \in (0, 1)$ for nonelite households and $\gamma_j = \gamma_E \in (0, 1)$ for elite households once the guarding margin becomes relevant. Rivalry in extraction remains explicit. For any household

j, let A_{jt} denote the set of active attackers who target j in period t and let $a_{jt} = |A_{jt}|$. Let $\omega_{jt} = 1$ if household j is occupied because its owner stays home, hires a thief, or hires a guard, and let $\omega_{jt} = 0$ if the owner leaves home to raid personally or works away from home as a thief. Let $g_{jt} = 1$ if household j is guarded and zero otherwise. Conditional on being targeted, the probability that loot is successfully extracted from j is

$$\kappa_{jt} = \mathbf{1}\{\omega_{jt} = 0\} + \chi_t \mathbf{1}\{\omega_{jt} = 1,\ g_{jt} = 0\} + (1 - q_t)\chi_t \mathbf{1}\{\omega_{jt} = 1,\ g_{jt} = 1\}, \tag{3.6}$$

where $\chi_t \in (0, 1]$ is the vulnerability of an occupied but unguarded household and $q_t \in [0, 1]$ is guard effectiveness. Expected gross loot taken from j is therefore $\kappa_{jt}\Gamma_{jt}$, and if $a_{jt} \geq 1$ each attacker receives expected share $\kappa_{jt}\Gamma_{jt}/a_{jt}$. Congestion in the extension is thus handled in the same way as in the benchmark: several attackers can converge on one victim, but they divide one rival object.

Within each period, wealth w_t is publicly observed before any sufficiently wealthy household chooses one occupation from the set $\{\text{stay}, \text{self raid}, \text{offer theft contract}, \text{offer guard job}\}$. Any poor household that receives offers then chooses at most one among $\{\text{accept theft employment}, \text{accept guard employment}, \text{freelance raid}, \text{stay}\}$. Every active thief next chooses a target, either directly if he is freelance or from the target class specified by the patron if he is employed, after which success, loot division, wages, remittances, and punishment are realized and wealth is updated. The patronage and guarding results below refer to this extension stage game and to stationary Markov behavior in its repeated version.

Two target classes keep the direction of predation clear. Let T_t denote the set of vulnerable nonelite households available to a wealthy self raider, and let $H_t \subseteq T_t$ denote the richer and weakly protected part of that set that a patron can reach more effectively by directing a dependent thief. In the patronage regime the physical act of theft is therefore still aimed mainly at poor or near-poor households. The vertical transfer appears because contracted thieves remit part of the loot upward to the patron.

For a wealthy household that raids personally, let m_t denote the expected gross return from the best target in T_t under the success and congestion rule just stated. For the pairwise comparisons below it is

enough to write

$$m_t = \gamma \bar{w}_t^T, \tag{3.7}$$

where $\bar{w}_t^T$ is the mean wealth of the best available target class among nonelite households after accounting for the fact that concurrent raiders dilute one another's shares. If rich agent i raids personally, the expected one period payoff is

$$\pi_{it}^{\text{self}} = m_t - \rho w_{it} - c_T, \tag{3.8}$$

where ρw_{it} is the expected value of leaving wealth exposed at home and c_T is direct theft cost. Staying home without contracting yields the normalized payoff $\pi_{it}^{\text{stay}} = 0$.

Delegation alters both the target set and the division of returns. A theft contract offered by patron i to poor worker j specifies a fixed payment s_{ijt}, a thief loot share $\alpha_{ijt} \in [0, 1]$, and access to the target class H_t. Let $\phi_t \geq 1$ summarize the patron's targeting advantage inside that class. Expected gross delegated loot is then

$$L_t = \phi_t \gamma \bar{w}_t^H, \tag{3.9}$$

where $\bar{w}_t^H$ is mean wealth in H_t. If the worker rejects the offer, he can freelance on the remaining vulnerable poor. Let o_{jt} denote that outside option:

$$o_{jt} = \lambda_t \gamma \bar{w}_t^P - c_T, \qquad \lambda_t \in (0, 1), \tag{3.10}$$

where $\bar{w}_t^P$ is mean poor wealth and λ_t summarizes the weaker targets and heavier congestion faced by freelancers. Worker j accepts the patron's offer if

$$s_{ijt} + \alpha_{ijt} L_t \geq o_{jt}. \tag{3.11}$$

Once a contract is accepted, one contracted theft relation is formed in that period. The patron pays contracting cost $k > 0$, the worker raids within H_t, the patron remains occupied at home, and realized

loot is divided according to the contract. Writing expected net joint surplus as

$$\Sigma_t = L_t - k - o_{jt}, \tag{3.12}$$

and letting $\beta \in [0, 1]$ denote patron bargaining power, the contract allocates

$$s_{ijt} + \alpha_{ijt} L_t = o_{jt} + (1 - \beta)\Sigma_t \tag{3.13}$$

to the thief, leaving the patron with $\beta\Sigma_t$. The induced expected wealth update for the patron-worker pair is

$$\begin{aligned} w_{i,t+1} &= w_{it} - \ell_{it} + (1 - \alpha_{ijt})L_t - s_{ijt} - k, \\ w_{j,t+1} &= w_{jt} - \ell_{jt} + s_{ijt} + \alpha_{ijt} L_t - c_T, \end{aligned} \tag{3.14}$$

where ℓ_{it} and ℓ_{jt} denote realized incoming losses under the success and congestion rule in (3.6). Rich agents who raid personally, poor freelancers, and guards are governed by the same accounting: period wealth equals initial wealth minus incoming loss plus role-specific income minus the resource cost attached to that role.

Lemma 3.7 (Contract feasibility)**.** *There exists a patron-thief contract satisfying the thief's participation constraint if and only if*

$$L_t - o_{jt} > k. \tag{3.15}$$

Proof. See Appendix A.1. A contract is feasible exactly when gross delegated loot exceeds the reservation utility of the thief plus the contracting cost of the patron. □

Condition (3.15) identifies the wedge that supports delegated predation. A patron gains nothing from hiring merely because wealth is high. Hiring becomes viable when access to H_t raises expected gross loot above what a freelancer can secure elsewhere and when congestion keeps the worker's outside option below that delegated return. Without that wedge, the contract market has no surplus to divide.

Proposition 3.8 (Strict patron surplus and occupational choice)**.** *Suppose Lemma 3.7 holds and the patron captures fraction β of the contract surplus. Then patron i's expected payoff from hiring thief j is*

$$\pi_{ijt}^{\text{hire}} = \beta \left(L_t - k - o_{jt} \right). \tag{3.16}$$

The rich agent strictly prefers hiring thief j to both abstention and personal theft whenever

$$\beta \left(L_t - k - o_{jt} \right) > \max \left\{ 0,\ m_t - \rho w_{it} - c_T \right\}. \tag{3.17}$$

Proof. See Appendix A.1. Equation (3.16) follows directly from the bargaining rule. Hiring dominates abstention because the right-hand side of (3.17) exceeds zero, and it dominates personal theft because the expected surplus from delegation exceeds the net return available to the rich agent who exposes wealth at home. □

Surplus arises because delegation improves access to targets and because the worker's reservation payoff remains below delegated gross returns. Better information, denser local knowledge, asymmetric bargaining power, or labor-market slack can each sustain that wedge. The poor workers who accept these offers remain poor enough that leaving home exposes little wealth, while the compensation implied by (3.13) still exceeds what freelance raiding would bring. The same pool of labor later becomes available for guarding.

Corollary 3.9 (Comparative statics of patronage)**.** *Fix w_{it} and suppose $L_t = \phi_t \gamma \bar{w}_t^H$ and $o_{jt} = \lambda_t \gamma \bar{w}_t^P - c_T$. Whenever (3.15) holds, patron payoff (3.16) is increasing in ϕ_t, decreasing in λ_t, decreasing in k, and decreasing in the attractiveness of personal theft only through the benchmark on the right-hand side of (3.17). In particular, increases in delegation quality and decreases in poor outside options expand the set of states in which rich agents choose patronage over both self-theft and abstention.*

Proof. See Appendix A.1. The derivative of (3.16) with respect to ϕ_t is positive, the derivatives with respect to λ_t and k are negative, and the comparison against personal theft depends on whether these

changes also widen the gap in (3.17). □

Accordingly, patronage expands when delegated targeting becomes more effective, contracts when the outside option of poor workers strengthens, and weakens when contracting becomes expensive.

3.5 The switch from patronage to guarding

Guarding is another action inside the same extension stage game. A household that hires a thief remains occupied but unguarded, so its own vulnerability is governed by the middle term in (3.6). A household that hires a guard remains occupied and guarded, so the success probability of an incoming elite raid falls to $(1 - q_t)\chi_t$. This distinction is what later separates patronage from protection: occupancy by itself no longer makes concentrated wealth safe once predation and retaliation have become selective.

Let elite household i offer a guard job at wage g to poor worker j. The worker accepts when the wage weakly exceeds both the freelance option and any noncriminal fallback:

$$g \geq \max\{o_{jt}, \underline{u}_{jt}\}, \tag{3.18}$$

where $\underline{u}_{jt}$ is the best noncriminal outside option. Once accepted, the worker remains at the elite house and does not raid in that period. The employer then receives the expected benefit of avoided elite loss:

$$\pi_{it}^{\text{guard}} = q_t \chi_t \gamma_E w_{it} - g, \tag{3.19}$$

where $\gamma_E w_{it}$ is the elite stock that would be exposed in a successful raid. Relative to one more patron-thief contract, guarding is optimal whenever the avoided expected elite loss exceeds the surplus from continued delegated predation:

$$q_t \chi_t \gamma_E w_{it} - g \geq \beta \left(L_t - k - o_{jt}\right). \tag{3.20}$$

Theorem 3.10 (Endogenous switch to guarding)**.** *Suppose guard labor is available at wage g satis-*

fying (3.18) *and patronage is initially feasible. Then guarding is optimal for elite agent* i *whenever*

$$w_{it} \geq \widehat{w}_t^G \equiv \frac{\beta\left(L_t - k - o_{jt}\right) + g}{q_t \chi_t \gamma_E}. \tag{3.21}$$

Moreover, $\widehat{w}_t^G$ *falls as the prey base among the poor is depleted, because* $L_t - o_{jt}$ *declines with* $\bar{w}_t^H$ *and* $\bar{w}_t^P$.

Proof. See Appendix A.1. Rearranging (3.20) yields the threshold. Depletion reduces the right-hand side of (3.20), lowering the elite wealth level at which guarding becomes privately optimal. □

Guard hiring does not appear immediately because elite wealth is not yet large enough and because patronage already keeps the employer at home while preserving some informal deterrence against attacks on patrons. Guards appear later because wealth concentration raises the loss at stake and because the surplus from one more delegated raid shrinks as poor targets are depleted. The direction of predation remains the same across that transition. During patronage, poor thieves are usually robbing poor households for wealthy principals. Once guarding dominates, part of that same labor pool is reassigned from predation to protection.

Corollary 3.11 (Comparative statics of guarding)**.** *The guarding threshold* $\widehat{w}_t^G$ *in* (3.21) *is decreasing in guard effectiveness* q_t*, decreasing in elite exposure* χ_t *and elite raid severity* γ_E*, increasing in the guard wage* g*, and increasing in the residual surplus from patronage* $\beta(L_t - k - o_{jt})$*. Guarding therefore becomes more likely when elite wealth is highly exposed and easily defended, and less likely when patronage remains highly profitable or guard labor becomes expensive.*

Proof. See Appendix A.1. The claims follow from direct differentiation of (3.21) with respect to the relevant parameters. □

Stronger guarding technology or greater elite exposure expands the region in which protection is privately optimal, while stronger patronage surplus or more expensive guard labor contracts that region. Policing emerges from the same strategic environment as patronage.

3.6 Welfare

The welfare criterion has to capture inequality and resource dissipation at the same time. An incomplete honest shock can move a society organized around reciprocal corruption toward more unequal and more resource-intensive regimes, while aggregate wealth alone remains uninformative because the benchmark conserves total wealth by construction. The analysis therefore uses discounted utilitarian welfare with explicit resource costs:

$$W_0 = \mathbb{E}_0 \sum_{t=0}^{\infty} \delta^t \mathcal{W}_t, \qquad \mathcal{W}_t = \sum_{i=1}^{N} u(w_{it}) - c_T n_t^S - k n_t^H - g n_t^G - \Psi_t, \tag{3.22}$$

where u is strictly increasing and strictly concave, n_t^S is the number of personal thieves, n_t^H is the number of active patron-thief contracts, n_t^G is the number of guards, and Ψ_t denotes expected punishment losses.

Proposition 3.12 (Redistribution, inequality, and resource cost)**.** *Suppose aggregate wealth $\sum_i w_{it}$ is fixed at date t and u is concave. Then any mean preserving spread of the wealth distribution weakly lowers $\sum_i u(w_{it})$. If, in addition, two regimes with the same aggregate wealth differ only in that one has weakly greater inequality and strictly greater values of n_t^S, n_t^H, n_t^G, or Ψ_t, then the latter regime yields strictly lower period welfare $\mathcal{W}_t$.*

Proof. See Appendix A.1. The first claim is Jensen's inequality. The second follows because the cost terms enter linearly and negatively in (3.22). □

A balanced corruption regime can already be normatively poor if theft effort and punishment risk consume resources, even when the distribution remains equal. Once the honest perturbation generates inequality and induces contracting and guarding costs, welfare can fall even if total wealth is approximately conserved in the short run. The relevant object is therefore distribution plus risk plus deadweight cost, and not redistribution alone.

Corollary 3.13 (Guarding may improve welfare relative to patronage)**.** *Fix the wealth distribution at date t. If adopting guarding lowers expected theft effort, contracting activity, and punishment*

losses by more than the resulting guard wage bill, then guarding raises $\mathcal{W}_t$ relative to patronage even though it need not restore the welfare of the original balanced benchmark.

Proof. See Appendix A.1. The difference in period welfare is the reduction in theft-related costs minus the new guarding expenditure. □

Organized enforcement can be privately attractive for elites and even socially preferable to active patronage on a utilitarian measure, while still leaving society worse off than the initial equal benchmark once one accounts for the path by which elite wealth was accumulated. The model therefore leaves room for a limited welfare improvement from policing relative to patronage without mistaking that improvement for a return to egalitarian order.

A balanced theft cycle is a coherent benchmark in the unit transfer stage game. A single permanent honest perturbation breaks that benchmark in the repeated game when wealth-dependent exposure is strong enough. After symmetry breaks, proportional theft together with weak poor outside options makes patronage feasible and profitable. Later, depletion of the prey base and concentration of elite wealth make guarding privately optimal. The analytical results identify those margins, and the simulations trace their timing, prevalence, and welfare consequences.

4 Simulation Design

Section 3 defines a unit transfer benchmark and an extension stage game with occupation choice, offers and acceptances, target selection, success probabilities, and rival loot. The simulations use those same primitives in a large stochastic population where repeated matching and endogenous wealth heterogeneity make closed form characterization impractical. The section records how those theoretical objects are implemented in code and which outcomes are stored for the figures and tables.

4.1 State variables, timing, and implementation

The simulation state in period t contains four objects: the wealth vector w_t, the role vector r_t, the patron-thief contract network, and the guard network. Roles take values in {honest, stay, freelance thief, hired thief, pat

with the understanding that an elite can also be a patron or a protected household once the relevant threshold is crossed. After wealth is observed, agents are classified into rich and elite sets by threshold rules based on current mean wealth, and the code then reproduces the extension stage game in the same order as Section 3. Rich households choose among staying, self raiding, and making theft offers. Elite households additionally compare continued patronage with guard hiring through (3.21). Poor households accept at most one offer or choose freelance predation or withdrawal. Active thieves then select targets, success is governed by (3.6), contractual payments and guard wages are applied, punishment is realized where relevant, and the next state is recorded.

Deterministic code handles the exact event accounting required by the first honest night, while stochastic code governs the richer transition once matching, targeting, and enforcement interact. The baseline seed is fixed at 613, every sweep uses reproducible seed offsets tied to the parameter grid, and every displayed result is generated from the same pipeline without hand-edited intermediates.

Event time is aligned on the first honest night, so the initial perturbation remains distinct from later adaptation. The baseline routine isolates the direct consequence of one honest household staying home. The later stochastic routines then measure what happens once wealth heterogeneity changes occupational choice, target selection, and protection decisions. Parameter changes can therefore be read through the channel they affect: amplification of the local asymmetry, the profitability of delegation, or the private value of guarding.

4.2 Mapping the theory into simulation rules

The benchmark simulation stays as close as possible to Theorem 3.6. Agents begin on a balanced cycle in the unit transfer environment, one agent is permanently honest, and on the first night the honest agent stays home while the old cycle is otherwise followed exactly. This reproduces the accounting in Lemma 3.4. The same collision rule used in the benchmark game is preserved, although no congestion occurs along the initial cycle because each target is chosen once. In later benchmark periods, agents compare their wealth to the cutoff in (3.5). Households below the threshold continue personal raiding along the cycle, while households above it stay home. Contracting, guarding, and stochastic targeting are absent

at this stage, so any departure from the balanced cycle is driven directly by the honest perturbation and the wealth-dependent exposure term.

Once proportional loot is introduced, role thresholds are defined relative to current mean wealth. An agent becomes rich when his wealth exceeds $\theta_R \bar{w}_t$, where $\bar{w}_t$ is the cross-sectional mean, and becomes elite when wealth exceeds the higher threshold $\theta_E \bar{w}_t$. These thresholds are reduced-form counterparts of the occupational comparisons in Section 3. Rich households are those for whom exposure makes personal raiding relatively unattractive, while elites are those for whom the stock of wealth at risk makes guarding salient. In the baseline calibration, the thresholds preserve event-time separation between the first honest shock and later institution building while still allowing the patronage regime to emerge within the simulation horizon.

Rich households are sorted by wealth, and candidate thieves are drawn from the poorest available agents who are neither permanently honest nor already employed. Sorting employers and matching them to the poorest available workers is the computational analogue of the offer stage in the extension game. A contract is formed only if the feasibility condition in Lemma 3.7 holds with current state variables. When a contract is feasible, the simulation applies the bargaining rule in (3.13): the thief receives his outside option plus the share of residual surplus not captured by the patron, and the patron receives the rest. The hired thief then targets households in the high value poor set H_t, while freelancers target vulnerable households by expected loot. Rival loot remains one object here as well. When several freelancers or hired thieves converge on the same house, the code applies the same congestion logic as (3.6) and (3.14): attackers draw from one vulnerable stock, so each additional attacker lowers the expected return of the others.

The target rule keeps the direction of predation explicit. In the patronage regime, poor thieves usually rob poor households for rich patrons. Physical theft remains largely horizontal among the nonelite, while the net transfer is vertical because part of the loot moves upward through the contract. The candidate target set excludes the patron and currently protected elite households. The patron's house remains occupied in the patronage phase, and betrayal risk enters only through the residual elite vulnerability summarized by χ_t once patronage begins to unravel.

Elites then face a second occupational comparison, namely whether the expected surplus from another patron-thief contract is smaller than the expected value of avoided elite loss from a guard. If inequality has risen and the prey base among the poor has thinned, the guarding condition in (3.21) is satisfied for some elite households. Guards are again recruited from the poorest available agents, but only if the guard wage exceeds their best alternative. Once hired, a guard reduces the success probability of raids on the protected house and introduces a punishment probability if an attempted elite raid fails. This gives the simulation an explicit mechanism by which formal protection suppresses open predation without removing the underlying population willing to steal.

Across the simulation, chronology runs through three stages that match the theory. The first stage is the deterministic first-night perturbation in the unit transfer benchmark. The second is the patronage stage in the proportional environment, where class differentiation and delegated predation emerge. The third is the enforcement stage, where some elites replace delegated predation with guards and punishment. Endogenous threshold crossings generate that chronology, so calendar dates do not drive the switch and transition timing remains an outcome of wealth dispersion, target depletion, outside options, and enforcement effectiveness.

4.3 Baseline calibration

Baseline parameter values are chosen in the internal units of the model and are not estimated from a specific dataset. The number of agents is set to $N = 60$, which is large enough to generate meaningful cross-sectional dynamics and small enough to keep network visualizations interpretable. Initial wealth is normalized to one for every agent. The proportional theft intensity is $\gamma = 0.35$, so a successful raid extracts a substantial but not total fraction of victim wealth. The rich threshold is $\theta_R = 1.15$ times current mean wealth, while the elite threshold is $\theta_E = 2.10$ times current mean wealth. These values allow early windfalls to matter without producing an elite stratum immediately.

The remaining parameters pin down the costs and comparative advantages of the occupational choices. The cost of theft effort is $c_T = 0.02$. The contracting cost is $k = 0.03$. The baseline guard wage is $g = 0.08$. The target thief share under a contract is $\alpha = 0.45$, while the fixed wage

Table 2: Baseline simulation parameters. The values define the benchmark environment used in the main figures, while the parameter sweeps vary the key margins systematically around this baseline.

Symbol	Value	Description
N	60	Number of agents
γ	0.350	Proportional theft intensity
$\bar{w}$	1	Initial wealth per agent
θ_R	1.150	Rich threshold multiplier
θ_E	2.100	Elite threshold multiplier
s	0.030	Base contract wage
α	0.450	Thief loot share under contract
κ	0.030	Contracting cost
g	0.080	Guard wage
q	0.980	Guard effectiveness

component is calibrated at $s = 0.03$. The success probability of raiding an occupied but unguarded household is low but positive, and the success probability against a guarded elite household is lower still. Punishment after a failed guarded raid is likely but not certain. In words, the calibration assumes a society in which unprotected wealth is vulnerable, protection is costly, and the poor can be induced to work as thieves or guards because their outside options are weak. Table 2 reports the core baseline values used in the main figures.

Parameters that govern the first night logic mirror the benchmark theory. Parameters that govern patronage and enforcement are chosen so that each regime can emerge for reasons consistent with the propositions in Section 3. The baseline outside option of poor thieves is low enough that patronage can be strictly profitable, but not so low that every rich agent hires immediately. Guard effectiveness is high enough that protection can dominate once wealth is sufficiently concentrated, but not so high that guards appear before a meaningful patronage phase.

The baseline path is deliberately moderate: inequality rises without exploding instantly, guards appear only after a visible patronage phase, and collapse remains possible without becoming inevitable. A more aggressive calibration would produce sharper pictures, but it would also obscure whether the mechanisms emerge from the stated thresholds or from extreme parameter values.

4.4 Parameter sweeps and regime classification

Systematic parameter sweeps then vary the margins highlighted in the theory. Theft intensity γ is varied to measure how sensitive class formation is to the fraction of wealth that can be appropriated in a single raid. Exposure risk is varied through the success probability of attacks on occupied but unguarded households, which determines how costly it is for a rich household to rely on occupancy alone. Contracting cost k and patron bargaining power β are varied to map the feasibility and profitability of delegation. Guard effectiveness and punishment severity are varied to locate the transition from patronage to protection. The outside option of poor thieves is varied through the congestion parameter λ_t and the availability of guard employment. Finally, the size of the initial honest set is varied so that the simulations can distinguish a single local perturbation from a broader coordinated honesty shock.

Each period in each simulation path is classified into one of six regimes: balanced cycle, transition, patronage, guard regime, coexistence, or collapse. The balanced-cycle label is reserved for periods in which almost all agents continue the reciprocal pattern and inequality remains negligible. Transition denotes periods after the first honest disturbance but before patronage or guarding has become dominant. Patronage denotes periods with active patron-thief contracts and no meaningful guard presence. Guard regime denotes periods in which protected elites and guards are established at scale. Coexistence denotes periods in which patronage and guarding coexist in nontrivial numbers. Collapse denotes periods in which the poor prey base is so depleted that predatory and protective roles no longer reproduce the earlier structure. These labels are not rhetorical. They are deterministic functions of role counts, inequality thresholds, and guard presence stored in the simulation output. The phase diagram in Section 5 is built directly from these classifications.

Fixed coding rules determine regime classification. Regimes are not inferred from unsupervised clustering, and paths are not relabeled ex post to fit the figures. The same classification rule is imposed across the entire parameter grid, which is necessary if a phase diagram is to carry comparative meaning.

4.5 Recorded outcomes

For each path the code stores agent level wealth, role histories, employer-worker links, target links, guard assignments, realized and expected welfare components, inequality measures, and first entry times into patronage and guard states. These objects support the figures that the theory alone cannot provide: event time wealth trajectories by eventual role, distributional heatmaps and quantile ribbons, Lorenz curves at key dates, phase diagrams across parameter pairs, and network snapshots of theft, patronage, and guarding. The same stored objects support the tabulated summaries of first patron entry, first guard entry, final inequality, and dominant long run regime.

Recorded outcomes answer a limited set of analytical questions tied to the theory: how the first honest perturbation propagates through the wealth distribution, when delegated predation becomes organizationally stable, and when organized protection replaces or disciplines predation. Event time plots establish the timeline. Distributional plots establish stratification. Comparative statics show that patronage and guarding do not depend on one baseline path. Welfare decompositions track what is lost and what, at times, is partially recovered as regimes change.

5 Simulation Results

5.1 The first honest night

Figures 1 and 2 begin with the exact accounting of the first honest night, because every later inequality result inherits its force from that local break. Figure 1 traces the honest agent, the predecessor, the successor, and the remaining population in event time. The pattern is exactly the one derived in Lemma 3.4. The honest agent does not suffer an immediate loss because he is at home on the first night. The predecessor who attempts to rob that occupied house receives no loot and is still robbed by his own predecessor, so his wealth falls. The successor, who would otherwise have been robbed by the honest agent, keeps the proceeds of his own raid and receives the first windfall. The initial rupture therefore takes the form of a precise redistribution within a three agent neighborhood of the original cycle.

Figure 2 compresses the same event into a cleaner accounting display. The honest agent is protected

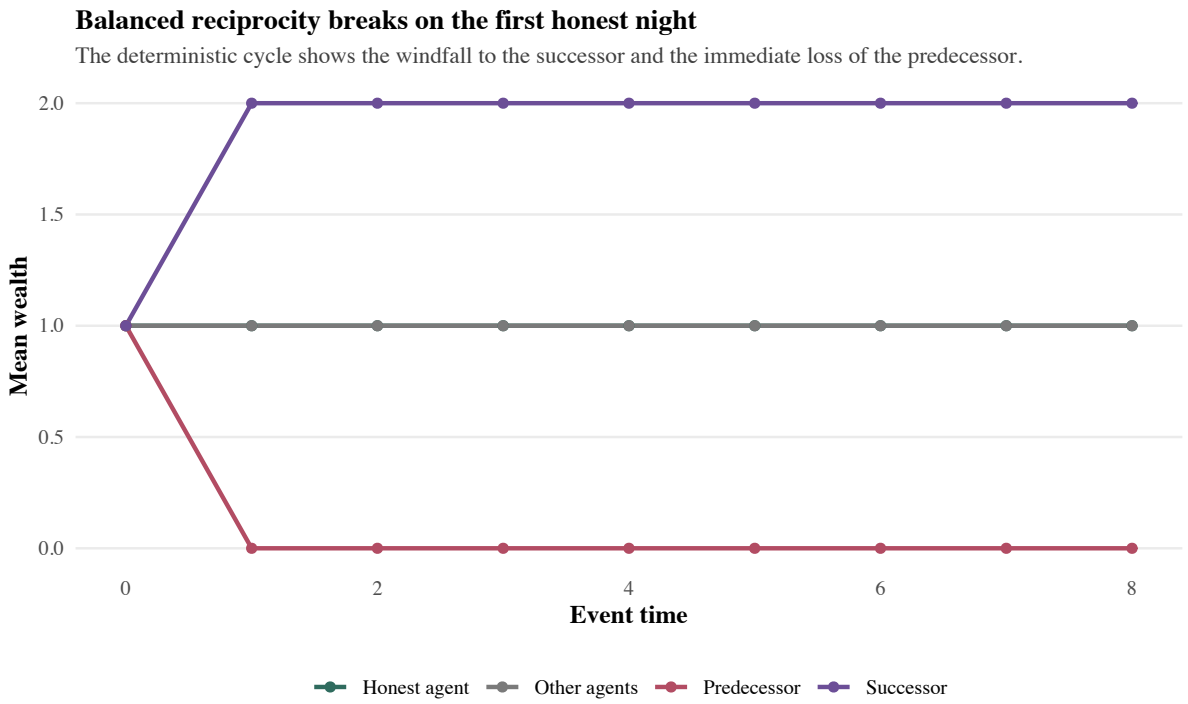

Figure 1: Balanced reciprocity breaks on the first honest night. The successor of the honest agent receives an immediate windfall, while the predecessor who attempts to rob the honest house returns empty handed. This local asymmetry starts the transition.

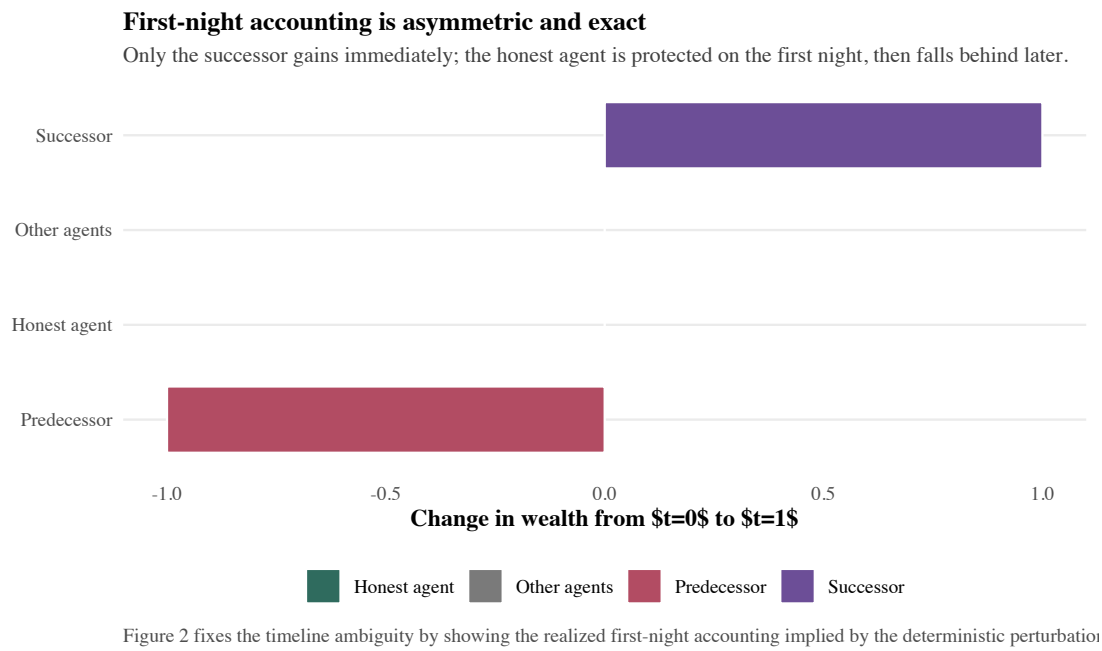

Figure 2: First night accounting. The honest agent is protected on the first night and is robbed only later in the transition, so the winner and loser on that night are the successor and predecessor in the original cycle. This timeline underlies the instability theorem.

on the first night, the successor receives the first windfall, and the predecessor is the first loser. Theorem 3.6 uses exactly that chronology through the local state change that places the successor at $\bar{w} + \tau$ and compares that wealth to the cutoff in (3.5). A different first-night chronology would imply a different state and a different continuation result.

The deterministic benchmark leaves no room to attribute the departure from the original cycle to chance. The simulation is implementing the same occupancy rule, the same first-night accounting, and the same cutoff comparison used in the theorem. Once successor wealth exceeds (3.5), staying home dominates further personal raiding, and the event-time display records the realized implication of that threshold comparison.

5.2 From local asymmetry to class formation

Longer horizon simulations show how the local state change created by Lemma 3.4 becomes a persistent class structure once agents are sorted by the payoff comparisons in Section 3. Figure 3 groups agents by the highest institutional role they reach during the transition and traces the corresponding

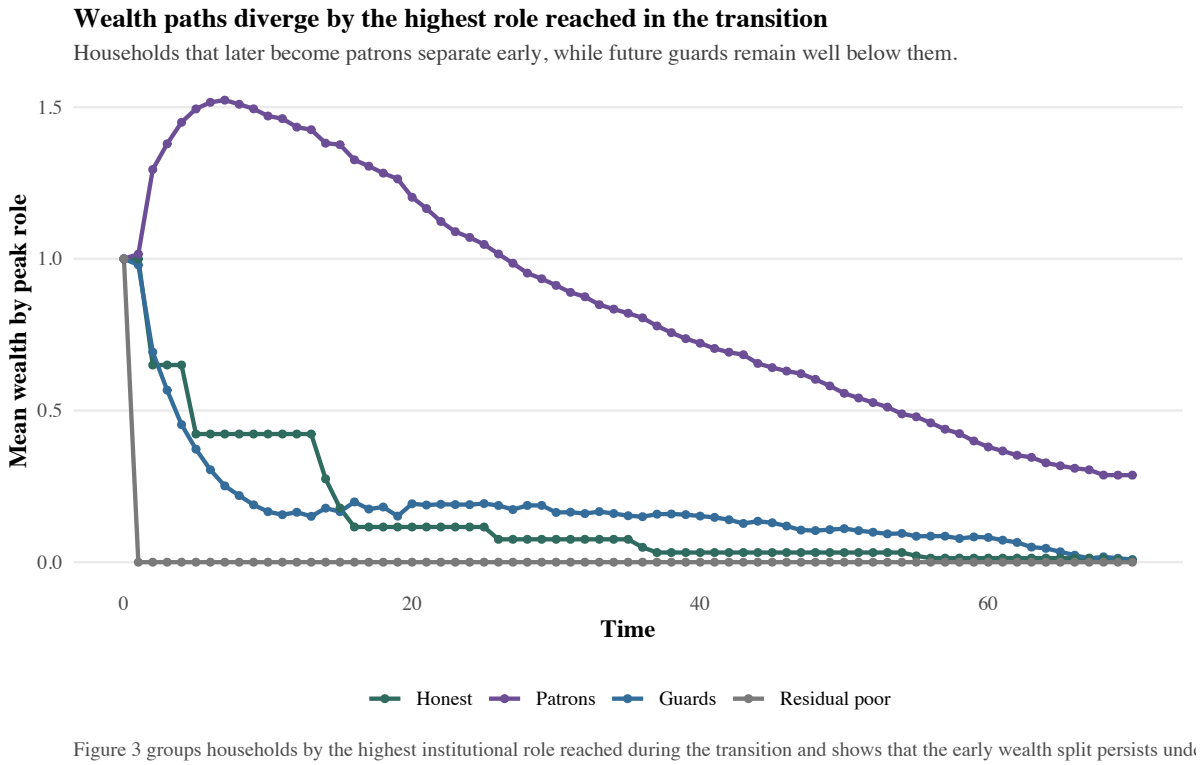

Figure 3: Event time wealth trajectories by peak role in the transition. Agents who ever become patrons separate quickly from future guards and the residual poor. Class differentiation appears early in the transition and does not depend on the terminal date.

wealth paths in event time, a grouping that preserves the early separation because many agents later withdraw into staying or inactivity. Agents who later become patrons separate first, agents who later serve as guards remain below them, and the honest household falls away sharply. This is the simulated counterpart of the exposure logic in (3.4): early windfall recipients stop exposing wealth sooner than the rest of the population, and later organizational roles inherit that initial gap rather than creating it from nothing.

Figure 4 shows the same process in cross section. The heatmap and quantile ribbons reveal a distribution that rapidly loses its middle. In early periods the support remains narrow because the honest perturbation is still local. A few periods later the upper tail thickens while the lower quantiles bend downward as repeated raids and weak outside options trap a growing mass of agents in the poor set. The widening is asymmetric for reasons already built into the theory. The upper tail is carried by agents whose wealth makes personal exposure costly and who later capture contractual surplus through

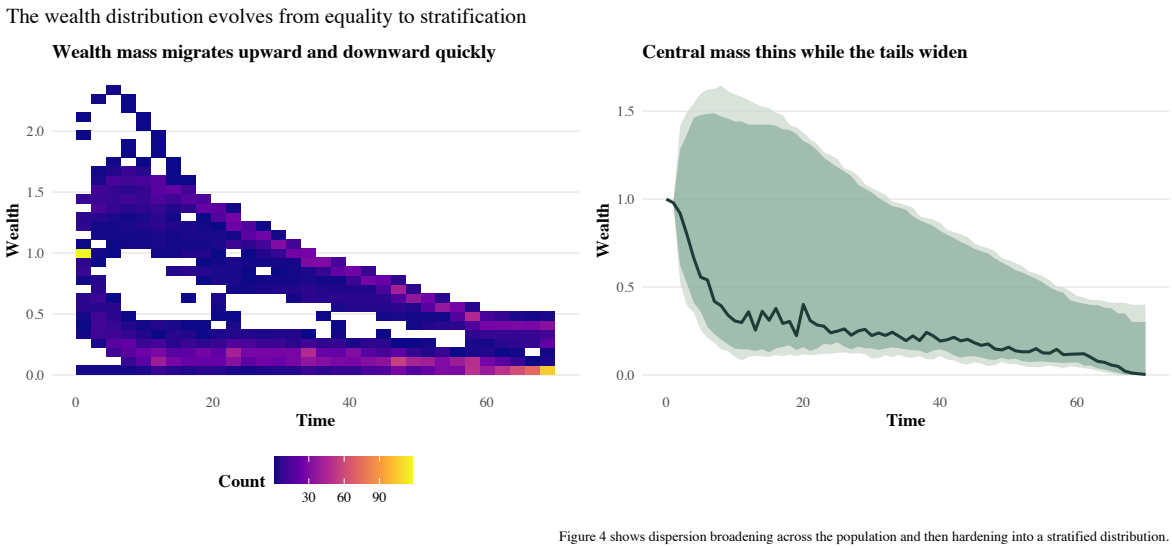

Figure 4: Evolution of the wealth distribution. The heatmap and quantile bands show the rapid thinning of the middle and the thickening of both tails. The honest perturbation produces durable stratification instead of a temporary reshuffle.

(3.16); the lower tail is carried by agents whose outside options remain close to (3.10) and whose wealth leaves them exposed to repeated predation. The cross section is therefore the distributional footprint of the occupational margins derived earlier.

Figure 5 translates that cross-sectional movement into conventional distributional statistics. The Gini coefficient rises sharply from its benchmark value near zero, the top decile wealth share increases in tandem, and Lorenz curves at selected dates move steadily away from the forty-five-degree line. These statistics do not establish a new mechanism beyond the one already derived. They show instead that the threshold logic of the model generates a large and persistent distributional divergence in the baseline calibration, not just a few isolated pathwise winners and losers.

Taken together, Figures 3 through 5 show the same mechanism at three levels. The deterministic shock creates a local state difference, the cutoff in (3.5) turns that difference into differentiated behavior, and the proportional environment amplifies the split because wealthy households cease to expose their own stocks while poor households remain available for predatory labor. The figures supply timing and magnitude; the underlying margins are already in the model.

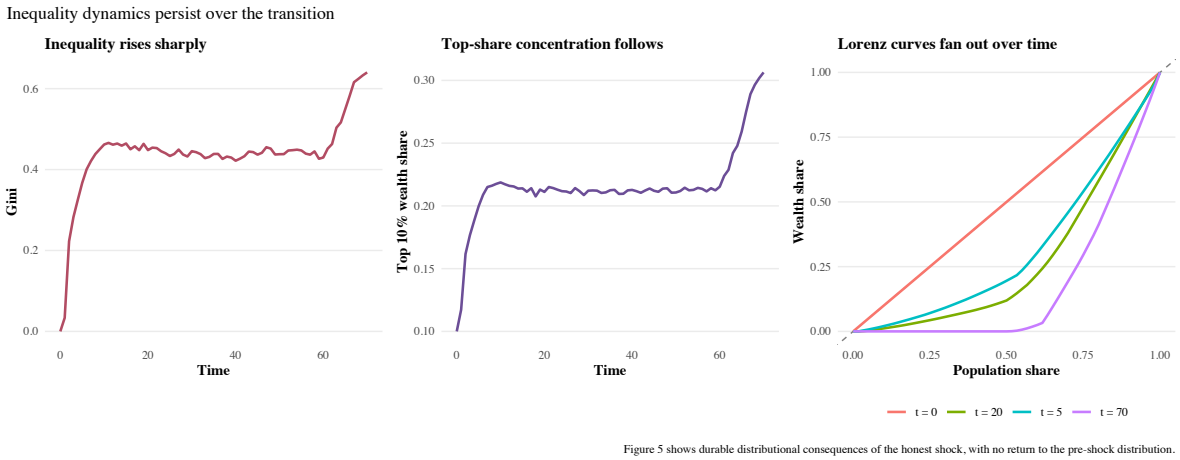

Figure 5: Inequality dynamics. The Gini coefficient, top decile wealth share, and Lorenz curves all move sharply away from the egalitarian benchmark. The transition has strong and persistent distributional consequences.

5.3 Patronage and the surplus from delegation

Across an interpretable range of parameters, rich households hire poor thieves only when the contract yields the surplus described by (3.12) and Proposition 3.8. Figure 6 examines that condition directly. The left panel varies the targeting advantage of delegation. The right panel traces compensation and the outside option as worker slackness changes. Patron surplus rises with the first margin and falls as the second margin strengthens, exactly as (3.16) and Corollary 3.9 imply. Delegated theft that is better targeted than freelancing leaves room to satisfy participation and still retain a positive residual. A stronger outside option narrows that residual and can eliminate it.

Delegation becomes profitable only because the environment generates a wedge between delegated returns and the outside option of the thief. If the thief could obtain alone the same gross loot available under delegation, and if the patron had neither targeting advantage nor bargaining power, (3.15) would fail or (3.17) would collapse to equality. The contracting region in the simulations is anchored in that explicit surplus condition, not in a looser story about hierarchy.

Patronage need not be monotone in every parameter that raises expected loot. When theft intensity becomes too large, the prey base can be depleted quickly, which shortens the durability of the contract regime even when delegation is initially attractive. Proposition 3.8 gives the static dominance condition; the simulations trace its interaction with depletion, competition, and regime duration.

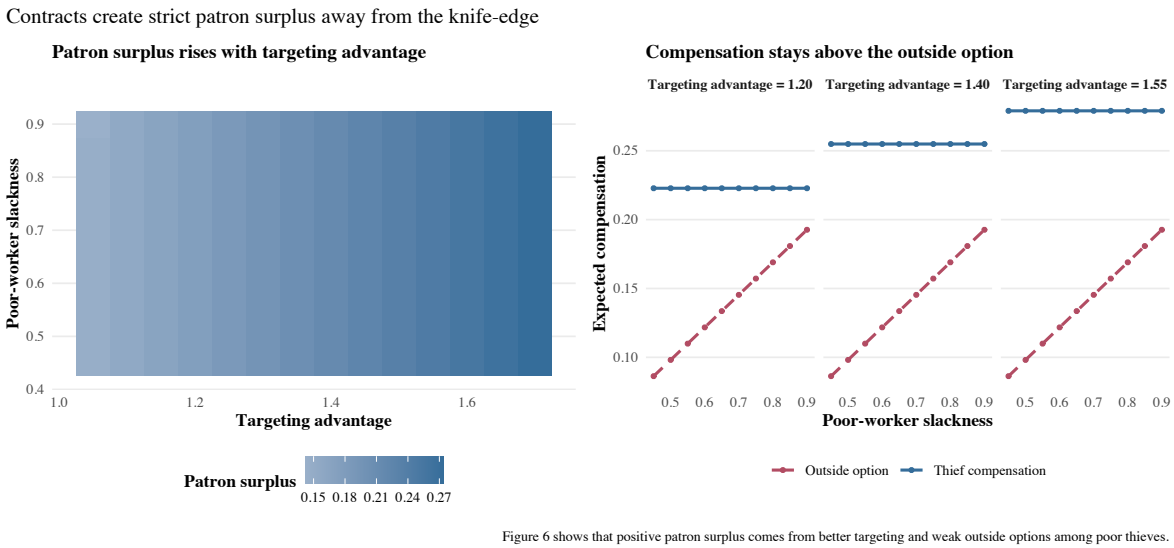

Figure 6: Contracting comparative statics. Patron surplus rises when delegated theft has a stronger targeting advantage and poor outside options are weaker. The source of patron surplus is explicit, and the earlier zero profit knife edge disappears.

5.4 From patronage to guarding

Figure 7 maps the dominant medium-run regime over the parameter plane spanned by theft intensity and enforcement effectiveness. Low theft intensity combined with weak enforcement tends to preserve a long transition because the returns to both predation and protection remain limited. As theft intensity rises and enforcement becomes more effective, the dominant regime shifts toward coexistence and guarding. Collapse appears where predation is intense enough and protection weak enough that the social base is depleted before a stable protected elite can consolidate. Equation (3.21) gives the organizing comparison: higher enforcement effectiveness lowers the private threshold for protection, while higher theft intensity both raises the stakes of exposure and accelerates depletion.

On this parameter plane, patronage appears mainly as a transitional corridor and only infrequently as the dominant medium-run regime. The diagram is therefore populated mostly by transition, coexistence, guard, and collapse regions, even though patronage remains visible in the event-time paths, network snapshots, and contracting comparative statics. Delegation organizes the middle of the transition without necessarily becoming the dominant late institutional form once (3.21) is crossed or depletion becomes severe.

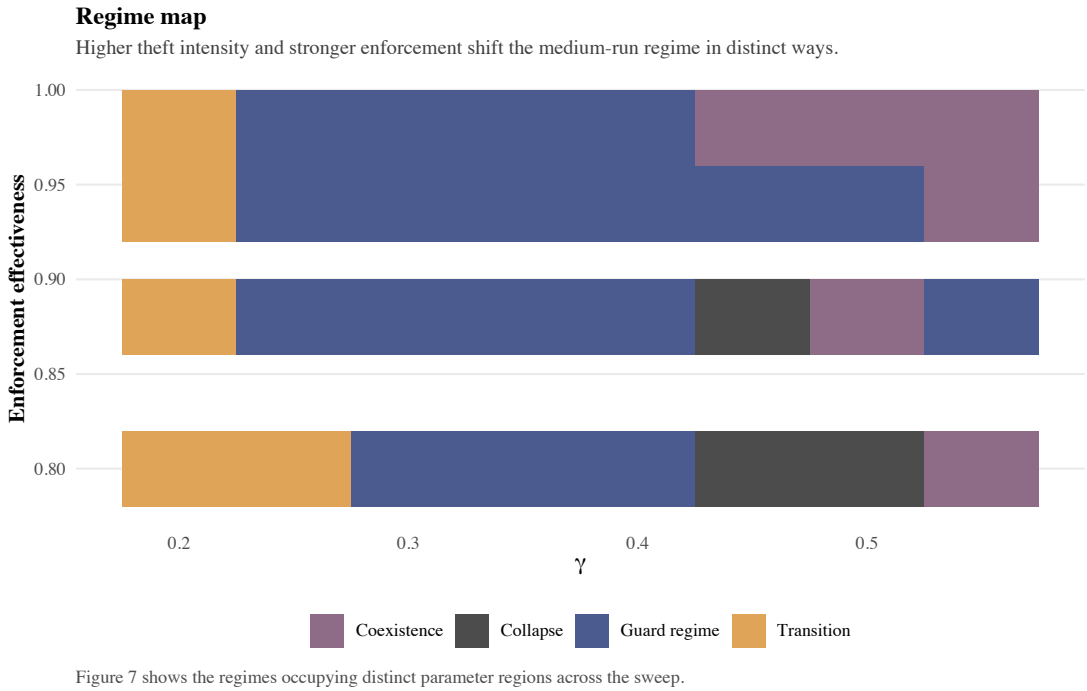

Figure 7: Regime phase diagram. Different combinations of theft intensity and enforcement effectiveness lead to distinct medium run transition, coexistence, guard, and collapse regions. The successor regimes occupy distinct and interpretable parts of the parameter space.

Figure 8 restores time to the same regime logic. The share of agents engaged in direct or freelance theft falls after the first perturbation. Patrons and hired thieves rise next. Guards appear later and remain active through much of the enforcement phase, even though their share is not monotone over time. The plot is useful precisely because each movement corresponds to a formal object already in the theory. The early decline in universal theft reflects Theorem 3.6; the rise in patrons and hired thieves reflects Proposition 3.8; the later appearance of guarding reflects Theorem 3.10.

Figure 9 separates the part of the transition pinned down by chronology from the part left open by later state evolution. Under the simulation timing, the first opportunity to hire arrives immediately after the deterministic honest shock, and every run generates at least one patron at that first post-shock contracting round, so a raw first-entry statistic for patronage carries little information. The left panel therefore measures the scale of the patronage complex at time 3, after the first hiring cascade has unfolded, while the right panel records the date at which guards reach five percent of the population. Early patronage is immediate under this timing rule; the movement into guarding is later and depends

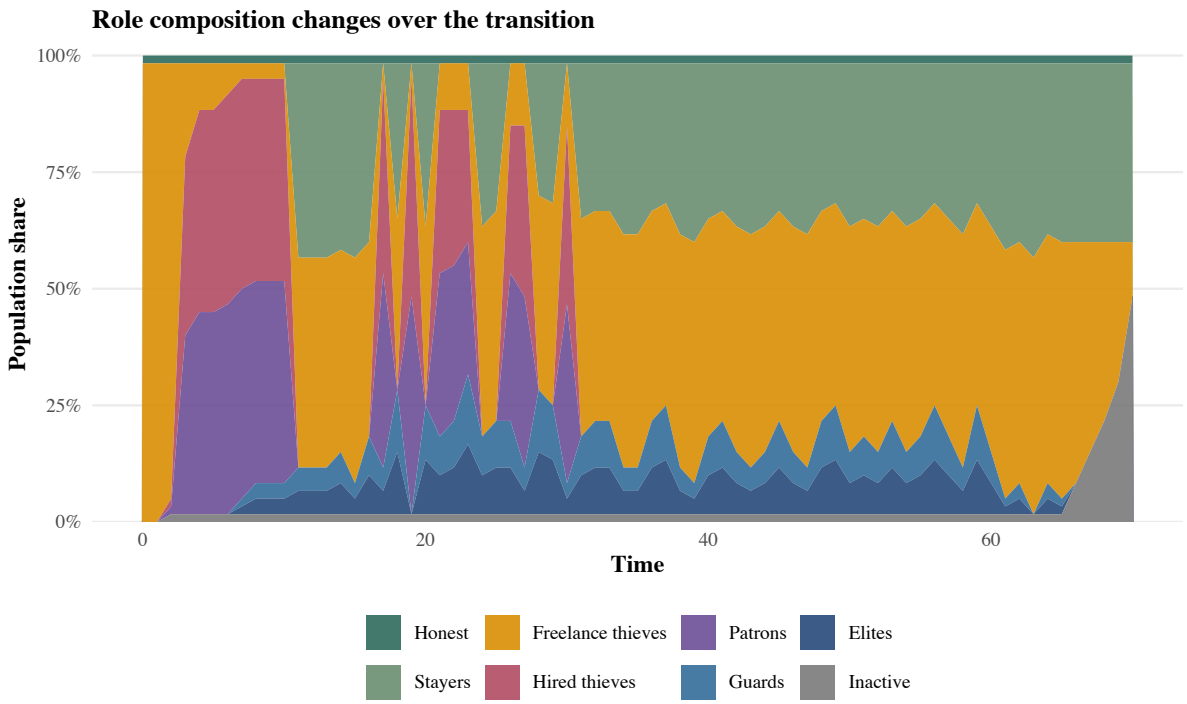

Figure 8: Role shares over time. Universal theft gives way first to patronage and later to selective protection, with guarding remaining present through much of the enforcement phase. The social hierarchy reorganizes endogenously after the first honest shock.

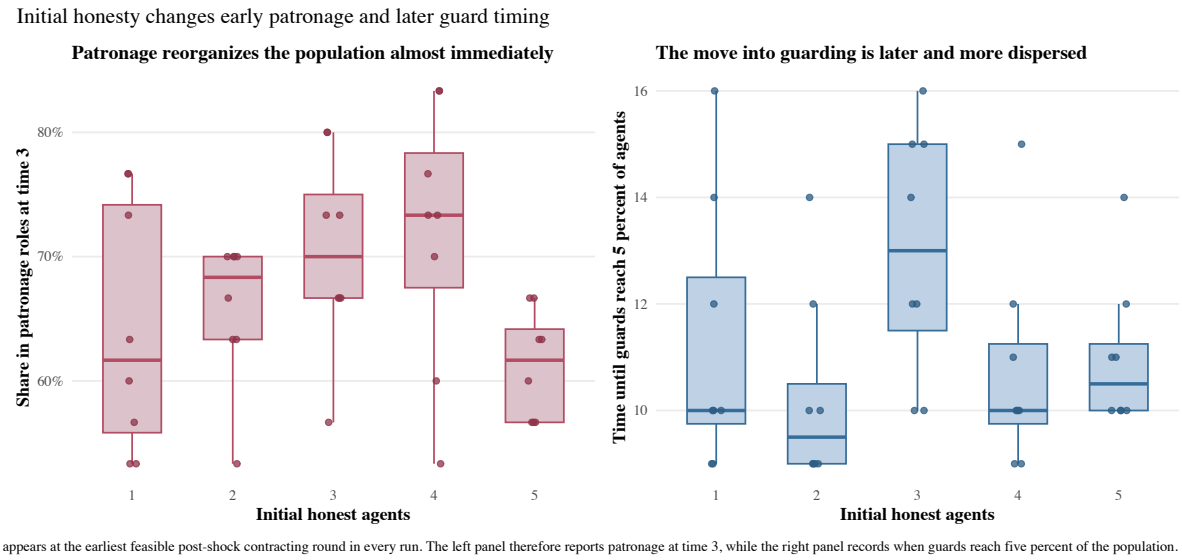

Figure 9: Early patronage and later guard timing. The first patron appears at the earliest feasible post-shock contracting round in every run, so the left panel reports the share of patrons and hired thieves at time 3. The right panel reports the time at which guards reach five percent of the population. Organizational change is immediate after the honest shock, while movement into enforcement arrives later and with more dispersion.

on the delayed crossing of (3.21).

Coexistence reveals institutional layering instead of a synchronized regime switch. In those paths, some elites have crossed the guard threshold while others still rely on patronage, so different technologies of control operate at the same time in different parts of the wealth distribution. The move from patronage to guarding is therefore unequal and sequential because (3.21) is wealth-dependent.

Table 3 compresses the same sequence across the baseline perturbation, patronage, and enforcement simulations after the timing, regime, and occupational evidence is in place. The baseline perturbation is the least unequal of the three simulated environments. Patronage produces a larger rise in concentration because it adds a contract technology that channels loot upward. Enforcement then preserves and slightly intensifies that concentration. The table is therefore the compact numerical counterpart to the welfare logic in Proposition 3.12: distributional divergence begins when reciprocity breaks and later regimes magnify or lock in that divergence through contracting and protection.

Table 3: Simulation summary. The first honest perturbation produces a local distributional break, patronage transforms that break into durable stratification, and enforcement preserves or deepens concentrated wealth instead of reversing it.

Scenario	Gini	Top decile	First patron	First guard	Regime
Baseline perturbation	0.231	0.200	Not reached	Not reached	Transition
Patronage dynamics	0.575	0.296	2	Not reached	Transition
Enforcement dynamics	0.640	0.306	2	7	Guard regime

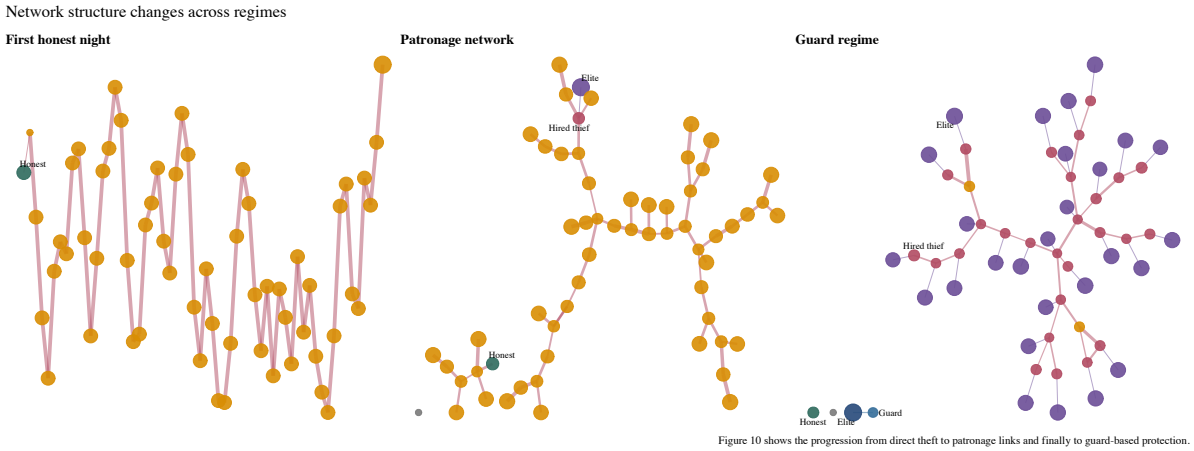

Figure 10: Network snapshots across regimes. Direct theft edges dominate at first, contract edges later connect patrons to thieves, and guard edges eventually organize elite protection. The network structure of predation and control changes with the regime.

5.5 Network reorganization and welfare

Figure 10 makes the organizational change visible in network form. On the first honest night the graph is still dominated by direct theft edges arranged around the original cycle. In the middle snapshot, contract links connect patrons to hired thieves and theft edges point from those thieves toward vulnerable poor households. In the later snapshot, guard links radiate from elite households while direct theft edges thin out. Proposition 3.8 and Theorem 3.10 imply that reorganization: direct predation gives way first to vertical contracting and later to selective protection.

Figure 11 reports the welfare consequences of the transition. Gross wealth and utilitarian welfare do not move together. Aggregate wealth declines because theft, punishment, and protection burn resources even when much of the interaction is redistributive in form. Welfare deteriorates more sharply because redistribution toward the top lowers the concave utility sum and because predation, contract-

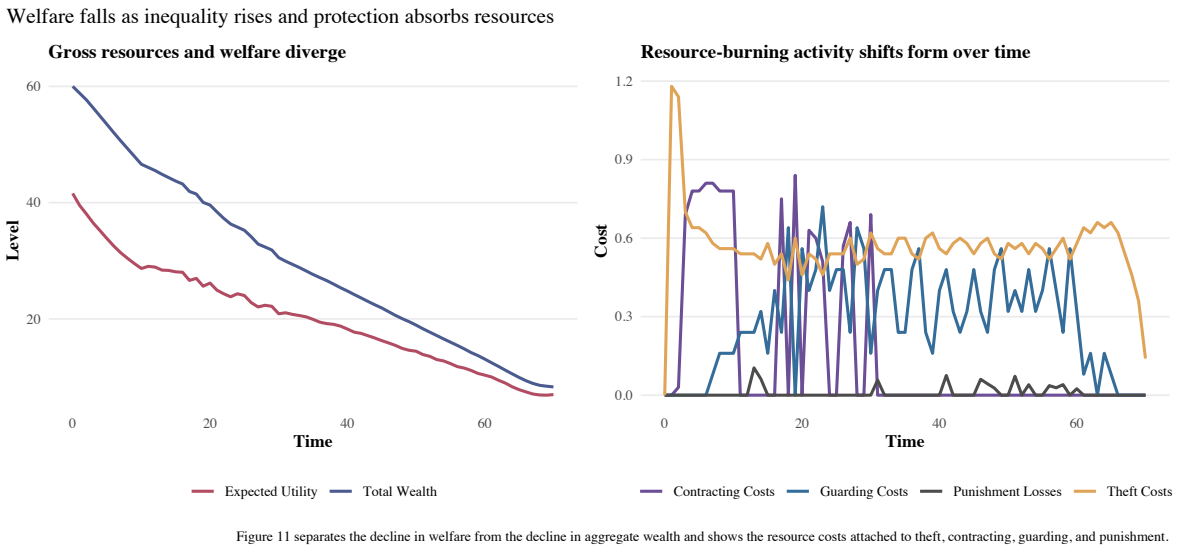

Figure 11: Welfare decomposition. Aggregate wealth and utilitarian welfare diverge because the transition adds inequality and resource burning activity in the form of theft, contracts, punishment, and guarding. The welfare mechanism appears directly in the decomposition.

ing, punishment, and guarding all consume resources. Proposition 3.12 formalizes that mechanism, and the figure decomposes it along one baseline path.

Because Figure 11 follows one baseline enforcement path rather than a comparison across parameter regions, its most direct contribution lies in the composition of resource use over time. Theft costs dominate early, contracting costs spike during patronage, and guarding costs remain persistent later on. Guard adoption changes the composition of resource loss without restoring the earlier welfare level.

5.6 Robustness

Figure 12 collects the main one-dimensional robustness exercises. Final welfare responds most strongly to theft intensity and occupied-house vulnerability in this sweep, while final inequality moves within a narrower band. Regime classification also changes little once the system is already in the region where guarding is attractive. Among the four margins shown, only the theft-intensity sweep changes the dominant regime, passing from transition to guard and then to coexistence as extraction becomes harsher. Figure 12 is therefore mainly about the strength of the mechanism, while Figure 7 remains the main source for regime geometry.

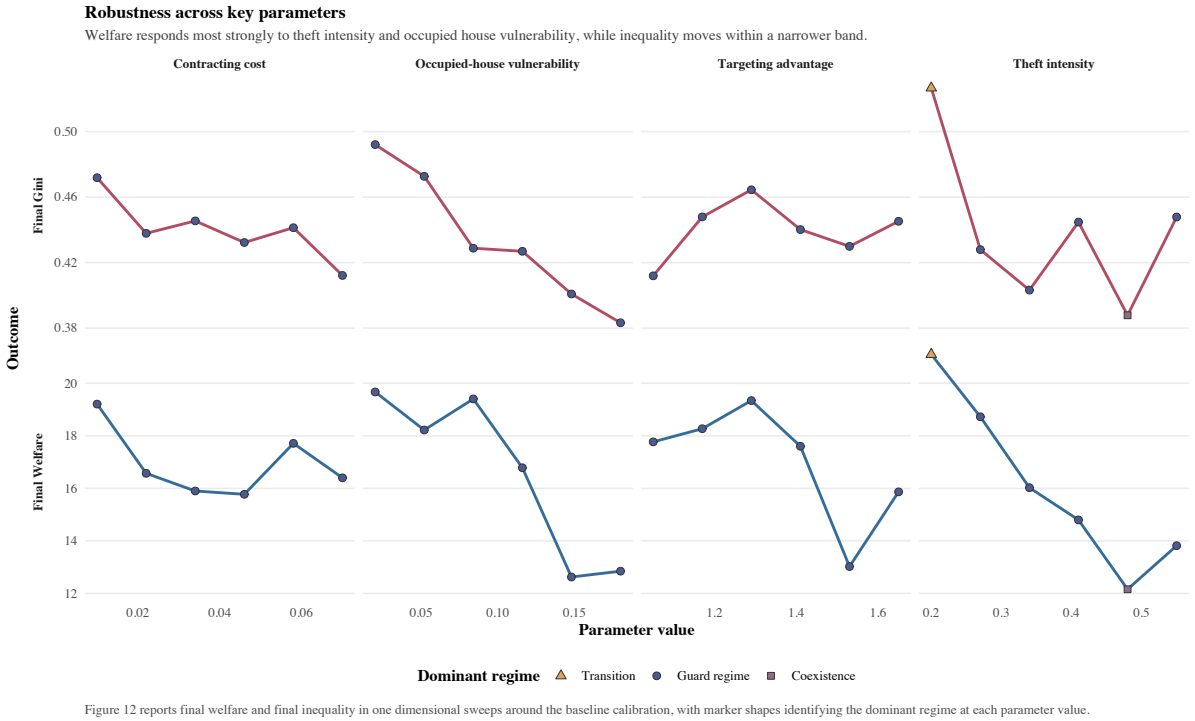

Figure 12: Robustness panels. The top row reports final inequality, the bottom row reports final welfare, and point markers identify the dominant regime reached at each parameter value. The mechanism survives beyond a single calibration, although some margins move welfare more than regime labels.

The failure cases identify the conditions under which the mechanism attenuates. High contracting costs and weak targeting advantage compress the surplus in (3.12), while low occupied-house vulnerability reduces the value of crossing the guarding threshold. The sweeps therefore locate the margins that mute the transition without suggesting that every one-dimensional variation redraws the institutional map.

Figure 7 is the more informative display for collapse. There the collapse cells do not mark a numerical artifact or a moral inversion of balanced reciprocity. They identify paths in which predation and protection together destroy the social base that made them profitable. Poor households become too depleted to sustain patronage, and inequality becomes so sharp that a small protected core cannot reorganize the rest into a stable order. This possibility prevents the model from telling a falsely deterministic story in which every corrupt equilibrium matures smoothly into guarded hierarchy.

6 Discussion and Conclusion

The opening order remains stable only in the narrow accounting sense captured by the unit transfer benchmark. Each household loses once and gains once, so equality survives even while predation is universal. Lemma 3.4 and Theorem 3.6 then show how little is required to destroy that balance. One permanently honest household blocks one incoming raid, leaves one successor unraided, and creates a local wealth difference large enough to matter once personal raiding also exposes current wealth. The later environments amplify that initial asymmetry through new margins. Proposition 3.8 requires a surplus from delegation over the worker outside option. Theorem 3.10 requires concentrated wealth and effective protection to make guarding privately preferable to one more round of delegated predation. Proposition 3.12 turns the same sequence into a welfare statement by combining concavity in wealth with the resource costs of theft, contracting, guarding, and punishment. The figures then supply timing, magnitude, and regime prevalence for those margins once many heterogeneous agents interact.

No full Markov perfect equilibrium is solved for the stochastic environment with matching, delegation, and guarding. Production is absent, so the model studies redistribution, dissipation, and protection of wealth rather than the emergence of a productive property-rights order. The final regime accordingly remains a protected unequal order built on stocks accumulated during the transition itself. The simulations make visible the state dependence that lies beyond the closed-form results. Event-time figures isolate the first-night perturbation and the resulting exposure threshold. Contracting figures identify the surplus region in which delegation is feasible. The phase diagram and robustness panels show where the guarding threshold is reached, where patronage remains transitional, and where depletion collapses the mechanism before a stable protected elite can consolidate. A broader coalition of honest households could generate a different path, and a model with production or a richer public-goods technology could generate a different endpoint.

Calvino's fable remains useful because it compresses the whole movement into one thought experiment: universal predation can look balanced from within, a local honest rupture can destroy that balance, and the order that follows can be quieter without becoming more equal. Each step now rests on an explicit game, a restricted equilibrium class, and a small set of threshold results. The simula-

tions fill in the transition path that those results leave open. Reciprocity, delegation, and protection are linked throughout by stated actions, stated payoffs, and stated conditions.

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Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Data Availability

All code, generated simulation outputs, and build scripts required to reproduce the figures and tables are archived on Zenodo at 10.5281/zenodo.19051109. The public GitHub repository remains available at github.com/cheikh-a/black_sheep_geb_submission_ready as the development mirror.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

Generative AI and AI-assisted tools were used under author supervision during revision of the manuscript to support language editing, code debugging, figure-production scripting, and consistency checks. All theoretical claims, modeling choices, simulations, interpretations, and editorial decisions were done, reviewed, and approved by the author, who takes full responsibility for the content of the article.

References

Daron Acemoglu and James A. Robinson. Persistence of power, elites, and institutions. *American Economic Review*, 98(1):267–293, 2008. doi: 10.1257/aer.98.1.267.

Daron Acemoglu, Davide Ticchi, and Andrea Vindigni. Emergence and persistence of inefficient states. *Journal of the European Economic Association*, 9(2):177–208, 2011. doi: 10.1111/j.1542-4774.2010.01008.x.

Jens Chr. Andvig and Karl Ove Moene. How corruption may corrupt. *Journal of Economic Behavior & Organization*, 13(1):63–76, 1990. doi: 10.1016/0167-2681(90)90053-G.

Pranab Bardhan. Corruption and development: A review of issues. *Journal of Economic Literature*, 35(3):1320–1346, 1997.

Gary S. Becker. Crime and punishment: An economic approach. *Journal of Political Economy*, 76 (2):169–217, 1968. doi: 10.1086/259394.

Gary S. Becker and George J. Stigler. Law enforcement, malfeasance, and compensation of enforcers. *Journal of Legal Studies*, 3(1):1–18, 1974. doi: 10.1086/467507.

Timothy Besley and Torsten Persson. The origins of state capacity: Property rights, taxation, and politics. *American Economic Review*, 99(4):1218–1244, 2009. doi: 10.1257/aer.99.4.1218.

Italo Calvino. The black sheep, 1993. URL https://granta.com/the-black-sheep/. English translation published in *Granta*, 46.

Harold Demsetz. Toward a theory of property rights. *American Economic Review*, 57(2):347–359, 1967.

Herschel I. Grossman and Minseong Kim. Swords or plowshares? a theory of the security of claims to property. *Journal of Political Economy*, 103(6):1275–1288, 1995. doi: 10.1086/601453.

Jack Hirshleifer. Anarchy and its breakdown. *Journal of Political Economy*, 103(1):26–52, 1995. doi: 10.1086/261974.

Douglass C. North, John Joseph Wallis, and Barry R. Weingast. *Violence and Social Orders: A Conceptual Framework for Interpreting Recorded Human History*. Cambridge University Press, Cambridge, 2009. doi: 10.1017/CBO9780511575839.

Trond E. Olsen and Gaute Torsvik. Collusion and renegotiation in hierarchies: A case of beneficial corruption. *International Economic Review*, 39(2):413–438, 1998. doi: 10.2307/2527300.

Mancur Olson. Dictatorship, democracy, and development. *American Political Science Review*, 87 (3):567–576, 1993. doi: 10.2307/2938736.

Lloyd Shapley and Herbert Scarf. On cores and indivisibility. *Journal of Mathematical Economics*, 1(1):23–37, 1974. doi: 10.1016/0304-4068(74)90033-0.

Andrei Shleifer and Robert W. Vishny. Corruption. *Quarterly Journal of Economics*, 108(3):599–617, 1993. doi: 10.2307/2118402.

Stergios Skaperdas. Cooperation, conflict, and power in the absence of property rights. *American Economic Review*, 82(4):720–739, 1992.

Jean Tirole. A theory of collective reputations (with applications to the persistence of corruption and to firm quality). *Review of Economic Studies*, 63(1):1–22, 1996. doi: 10.2307/2298112.

H. Peyton Young. The evolution of conventions. *Econometrica*, 61(1):57–84, 1993. doi: 10.2307/2951778.

A Appendix

A.1 Proofs

This appendix collects the proof details omitted from the main text. None of the omitted steps is especially difficult. The purpose of placing them here is simply to keep the main text readable while making the logical dependence of the results explicit. Throughout, the benchmark unit loot is $\tau = 1$ unless otherwise noted.

Proof of Lemma 3.2

Fix a balanced theft cycle σ and suppose every agent enters the period with wealth $\bar{w}$. By definition of a balanced cycle, each household is unoccupied and is targeted by exactly one raider, so $m_{jt} = 1$ for every j. Equation (3.1) therefore implies $\ell_{it}(a_t) = \tau$ and $g_{it}(a_t) = \tau$ for every agent i. Equation (3.2) becomes

$$w_{i,t+1} = \bar{w} - \tau + \tau = \bar{w}$$

for every agent i. Summing across agents yields

$$\sum_{i=1}^{N} w_{i,t+1} = \sum_{i=1}^{N} \bar{w} = N\bar{w} = \sum_{i=1}^{N} w_{it}.$$

Hence wealth is preserved both agent by agent and in aggregate. □

Proof of Proposition 3.3

Take an arbitrary agent i in a balanced theft cycle. If i follows the prescribed action, then i leaves home, loses τ to the unique incoming raid, and gains τ from the unique outgoing raid, finishing with wealth $\bar{w}$.

If i deviates to staying home, the incoming raid on i fails because the house is occupied, but i also gives up the outgoing loot. The deviating payoff is therefore still $\bar{w}$, so staying home is not strictly profitable.

If instead i deviates to another target $j \neq \sigma(i)$, then household j is already targeted by exactly one raider under the balanced cycle. The deviator therefore creates congestion at j and receives expected loot $\tau/2$ rather than τ. The incoming raid on i still succeeds because i remains away from home. The deviating payoff is then

$$\bar{w} - \tau + \frac{\tau}{2} = \bar{w} - \frac{\tau}{2} < \bar{w}.$$

Thus every retargeting deviation is strictly worse, while staying home is payoff equivalent. No unilateral deviation is strictly profitable, so the balanced cycle is a weak Nash equilibrium. □

Proof of Lemma 3.4

Let h be the permanently honest agent, $p = \sigma^{-1}(h)$ his predecessor in the original cycle, and $q = \sigma(h)$ his successor. At $t = 0$, agent h stays home. Because occupied houses cannot be burgled in the benchmark environment, the raid by p against h fails, so p gains no loot from his planned target. Agent h also undertakes no outgoing raid, so q is not robbed by h. All other agents follow the original cycle exactly.

Now compute end-of-night wealth. Agent h neither loses nor gains, so $w_{h1} = \bar{w}$. Agent p is still robbed by his own predecessor but gains nothing from his failed attempt on h, so $w_{p1} = \bar{w} - \tau$. Agent q still robs his own target but is no longer robbed by h, so $w_{q1} = \bar{w} + \tau$. Every other agent both loses τ and gains τ, hence remains at $\bar{w}$. This proves (3.3). □

Proof of Theorem 3.6

By Lemma 3.4, the successor q enters period $t = 1$ with wealth $w_{q1} = \bar{w} + \tau$. The expected one period net payoff from personal theft at wealth w is

$$\pi^S(w) = \tau - \rho w - c_T.$$

The payoff from staying home is normalized to zero. Therefore the agent strictly prefers staying home whenever $w > \widehat{w} = (\tau - c_T)/\rho$.

Assumption 3.5 states that $\bar{w} < \widehat{w} < \bar{w} + \tau$. The lower inequality ensures that at the symmetric benchmark wealth level $\bar{w}$ an agent weakly prefers theft or is indifferent, which is why the original cycle can be sustained before the honest perturbation. The strict upper inequality ensures that after the first-night windfall the successor's wealth lies strictly above the cutoff. Hence

$$\pi^S(w_{q1}) = \tau - \rho(\bar{w} + \tau) - c_T < \tau - \rho\widehat{w} - c_T = 0$$

so staying home is strictly better for the successor at state w_{q1}.

To prove the theorem as stated, suppose by contradiction that there exists a stationary Markov equilibrium preserving the original balanced theft pattern among all nonhonest agents. Then q must choose personal theft at state w_{q1}. But the threshold condition implies that staying home strictly dominates personal theft at exactly that state. This contradicts the alleged stationary continuation. Therefore no such stationary Markov equilibrium exists. □

Proof of Lemma 3.7

Fix a rich patron i and a candidate poor thief j. Let delegated gross loot be L_t, contracting cost k, and the outside option of the poor worker o_{jt}. A contract with terms (s_{ijt}, α_{ijt}) is feasible if the participation constraint of the thief holds:

$$s_{ijt} + \alpha_{ijt}L_t \geq o_{jt}.$$

The net return of the patron from the contract is

$$(1 - \alpha_{ijt})L_t - s_{ijt} - k = L_t - (s_{ijt} + \alpha_{ijt}L_t) - k.$$

For some contract to be mutually acceptable, there must exist a transfer to the thief that is at least o_{jt} and leaves the patron weakly nonnegative. This requires

$$o_{jt} \leq s_{ijt} + \alpha_{ijt}L_t \leq L_t - k.$$

Such an interval exists if and only if $L_t - k - o_{jt} \geq 0$, with strict inequality for strictly positive joint surplus. This is exactly condition (3.15). □

Proof of Proposition 3.8

Suppose (3.15) holds, so total joint surplus is

$$\Sigma_t = L_t - k - o_{jt} > 0.$$

Under the bargaining rule in (3.13), the thief receives

$$o_{jt} + (1 - \beta)\Sigma_t$$

in expected value, while the patron receives the residual

$$L_t - k - [o_{jt} + (1 - \beta)\Sigma_t] = L_t - k - o_{jt} - (1 - \beta)\Sigma_t = \beta\Sigma_t.$$

This proves (3.16). Hiring dominates abstention whenever $\beta\Sigma_t > 0$, and it dominates personal theft whenever

$$\beta\Sigma_t > m_t - \rho w_{it} - c_T.$$

Combining the two comparisons yields (3.17). □

Proof of Corollary 3.9

Using (3.16) together with $L_t = \phi_t \gamma \bar{w}_t^H$ and $o_{jt} = \lambda_t \gamma \bar{w}_t^P - c_T$, patron payoff can be written as

$$\pi_{ijt}^{\text{hire}} = \beta \left(\phi_t \gamma \bar{w}_t^H - k - \lambda_t \gamma \bar{w}_t^P + c_T \right).$$

Differentiating yields

$$\frac{\partial \pi_{ijt}^{\text{hire}}}{\partial \phi_t} = \beta\gamma\bar{w}_t^H > 0, \qquad \frac{\partial \pi_{ijt}^{\text{hire}}}{\partial \lambda_t} = -\beta\gamma\bar{w}_t^P < 0, \qquad \frac{\partial \pi_{ijt}^{\text{hire}}}{\partial k} = -\beta < 0.$$

Thus patron payoff rises with delegation quality and falls with poor outside options and contracting cost. Since patronage is chosen when (3.17) holds, any change that raises the left-hand side of that inequality, holding the right-hand side fixed, expands the set of states in which patronage dominates. □

Proof of Theorem 3.10

Elite agent i compares the payoff from guarding,

$$\pi_{it}^{\text{guard}} = q_t\chi_t\gamma_E w_{it} - g,$$

with the payoff from one more patron-thief contract,

$$\pi_{ijt}^{\text{hire}} = \beta\left(L_t - k - o_{jt}\right).$$

Guarding is optimal if and only if

$$q_t\chi_t\gamma_E w_{it} - g \geq \beta\left(L_t - k - o_{jt}\right),$$

which rearranges to

$$w_{it} \geq \frac{\beta\left(L_t - k - o_{jt}\right) + g}{q_t\chi_t\gamma_E} \equiv \widehat{w}_t^G.$$

This proves the threshold expression in (3.21). To show that the threshold falls as the poor prey base is depleted, note that both $L_t = \phi_t\gamma\bar{w}_t^H$ and $o_{jt} = \lambda_t\gamma\bar{w}_t^P - c_T$ depend positively on the wealth of the poor. Under the maintained assumption that delegated targeting advantage shrinks more slowly than the broad poor outside option, depletion lowers net delegated surplus $L_t - o_{jt}$ and hence lowers the threshold wealth level at which guarding dominates. □

Proof of Corollary 3.11

Write the guarding threshold as

$$\widehat{w}_t^G = \frac{\beta(L_t - k - o_{jt}) + g}{q_t \chi_t \gamma_E}.$$

Direct differentiation gives

$$\frac{\partial \widehat{w}_t^G}{\partial q_t} = -\frac{\beta(L_t - k - o_{jt}) + g}{q_t^2 \chi_t \gamma_E} < 0, \qquad \frac{\partial \widehat{w}_t^G}{\partial \chi_t} = -\frac{\beta(L_t - k - o_{jt}) + g}{q_t \chi_t^2 \gamma_E} < 0,$$

$$\frac{\partial \widehat{w}_t^G}{\partial \gamma_E} = -\frac{\beta(L_t - k - o_{jt}) + g}{q_t \chi_t \gamma_E^2} < 0, \qquad \frac{\partial \widehat{w}_t^G}{\partial g} = \frac{1}{q_t \chi_t \gamma_E} > 0.$$

Finally,

$$\frac{\partial \widehat{w}_t^G}{\partial \{\beta(L_t - k - o_{jt})\}} = \frac{1}{q_t \chi_t \gamma_E} > 0.$$

Hence stronger guarding technology or greater elite vulnerability lowers the wealth threshold for protection, while more profitable patronage or more expensive guard labor raises it. □

Proof of Proposition 3.12

Fix date t and suppose aggregate wealth $\sum_i w_{it}$ is fixed. Let u be concave. If one wealth vector is a mean preserving spread of another, then by the Hardy-Littlewood-Polya majorization theorem, or equivalently by Jensen's inequality applied to pairwise spreads, the more unequal vector yields a weakly smaller value of $\sum_i u(w_{it})$. This proves the first claim.

Now compare two regimes with the same aggregate wealth. If one regime has weakly greater inequality, then its utility sum is weakly lower. If, in addition, at least one of the cost terms n_t^S, n_t^H, n_t^G, or Ψ_t is strictly larger and none is smaller, then subtracting those costs from the already weakly smaller utility sum yields strictly lower period welfare $\mathcal{W}_t$. This proves the second claim. □

Proof of Corollary 3.13

Fix the wealth distribution. The difference between period welfare under guarding and period welfare under patronage is

$$\Delta\mathcal{W}_t = -gn_t^G - c_T n_t^{S,\text{guard}} - kn_t^{H,\text{guard}} - \Psi_t^{\text{guard}} + c_T n_t^{S,\text{pat}} + kn_t^{H,\text{pat}} + \Psi_t^{\text{pat}}.$$

If the reduction in theft effort cost, contracting cost, and punishment loss exceeds the new guard wage bill, then $\Delta\mathcal{W}_t > 0$. Since the wealth distribution is fixed by hypothesis, the utility term cancels. Hence guarding raises welfare relative to patronage under the stated condition. □

A.2 Alternative assumptions and scope

Several assumptions deserve comment because they determine both what the paper proves and what it leaves open. The first is the use of a unit transfer benchmark followed by a proportional transfer extension. A single proportional model could in principle have been used throughout. The difficulty is that the egalitarian balance at the beginning of the story by Calvino is most transparent when every successful raid transfers the same unit amount and every agent has exactly one incoming and one outgoing link. The benchmark therefore isolates the cycle cleanly. Once the paper turns to inequality and class formation, however, the unit transfer environment becomes too rigid because it cannot represent the idea that richer agents risk more by leaving home. The proportional extension is introduced at exactly that point and no earlier.

The second assumption concerns stationary Markov strategies in the honest-perturbation theorem. A reader might ask whether history-dependent punishments could sustain a one-honest continuation even after the first windfall. The answer may well be yes in some environments, but that is not the object of the theorem. The theorem isolates the local best-response logic that the simulation later implements. It says that once the wealth of the successor moves above the exposure threshold, the original cycle is not a stationary continuation. That is the appropriate statement for a paper interested in regime transition and not in the folk-theorem support of arbitrary histories.

The third assumption concerns the outside option of poor workers. The paper models it as weak because poor agents compete for a depleted set of vulnerable targets and because bargaining power is tilted toward patrons. One could instead model search frictions, repeated matching, or explicit auctioning of predatory labor. Any of those would add realism, but the key analytical requirement would remain the same: patronage must create a wedge between delegated gross returns and the best alternative available to poor workers. Without that wedge, the contract market has no surplus to divide.

The fourth assumption concerns guarding. In the model, guarding becomes attractive once occupancy alone no longer fully protects concentrated wealth after patronage weakens and poaching risk rises. This is summarized by the residual elite-risk term χ_t. A more elaborate microfoundation could model betrayal by former hired thieves, explicit coordination among freelancers, or the public-good aspect of policing. The reduced form used here is meant only to capture the fact that highly concentrated wealth becomes worth defending with a more dedicated technology than merely staying home.

A.3 Simulation implementation details

The algorithm below maps the formal objects into the computational steps that generate the figures and tables. The code remains the authoritative implementation, but the written version keeps the appendix in the same notation as the model and simulation sections.

1. Initialize N agents with common wealth $\bar{w}$, assign the balanced theft permutation σ, and designate the initial honest set. In the baseline path the honest set contains one permanent honest agent.

2. Run the deterministic benchmark perturbation. On the first night the honest agent stays home, all other agents follow the original cycle, and wealth is updated according to the accounting in Lemma 3.4.

3. In later benchmark periods, let each nonhonest agent compare current wealth to the threshold $\widehat{w}$ in (3.5). Agents above the threshold stay home; agents below it continue the cycle.

4. For the proportional transfer simulations, reinitialize from the common wealth state and apply the

first honest perturbation, then classify rich and elite agents each period using the threshold multipliers θ_R and θ_E.

5. Compute candidate personal theft returns for rich agents using (3.8). Compute candidate contract surplus using (3.12) and accept only contracts satisfying (3.15). Order patrons from richest to poorest and match them to the poorest available workers, which is the code-level implementation of the offer stage in the extension game.

6. Assign hired thieves to the high value poor target set H_t, freelancers to vulnerable households ranked by expected loot, and guards to elite households satisfying (3.21). Apply the guard acceptance condition (3.18).

7. Realize raids sequentially as the code-level counterpart to the rival-loot rule in (3.6). Each realized transfer is based on the victim's remaining wealth at the moment of attack, so collisions on one house generate diminishing returns within a period.

8. Apply wages, loot shares, contracting costs, guard wages, and punishment losses. Update the wealth vector and the role vector.

9. Compute inequality statistics, welfare components, first entry times into patronage and guarding, and the regime label for the period. Store these outputs before advancing to the next date.

10. Repeat until the simulation horizon is reached. Store processed data, figures, and tables through the reproducible build pipeline documented in the replication materials.

Three implementation choices are especially important. First, wealth changes are recorded in realized rather than merely expected values, but the occupational comparisons are based on expected payoff formulas that correspond to the theory. Second, occupied but unguarded houses are treated differently in the first night benchmark and the later stochastic simulations. In the benchmark they are perfectly safe because that is what the instability theorem requires. In the later simulations they are imperfectly safe because concentrated wealth eventually requires a dedicated protection technology.

Table 4: Dominant regime frequencies in the main parameter sweep. The table reports how often each regime is the dominant late period outcome over the grid used to produce the phase diagram.

Dominant regime	Frequency
Guard regime	18
Coexistence	6
Transition	5
Collapse	3

Third, the code distinguishes clearly between rich households hiring poor thieves to rob poor targets and elites later hiring poor guards to defend elite assets. This distinction was blurred in the earlier draft and is now maintained throughout.

A.4 Supplementary robustness and regime frequencies

The phase diagram in the main text summarizes the most informative parameter plane, but the code evaluates broader robustness variations. These include changes in the outside option parameter, the number of initially honest agents, the guard punishment probability, and the vulnerability of occupied but unguarded homes. The overall message is the same as in the main text: the mechanism is robust within a substantial region, but it does not survive when the analytical conditions behind the main propositions are switched off. If delegated theft has no surplus, patronage disappears. If guarding is ineffective or too expensive, elites continue with patronage or the system collapses. If theft intensity is too low, the society can remain close to the transition regime for long periods.

Table 4 reports the frequency with which each dominant regime appears in the main parameter sweep. These frequencies are not structural predictions, because they depend on the chosen grid. Their value is descriptive. They show that transition, coexistence, guarding, and collapse all appear in the sweep, which confirms that the model generates multiple successor regimes and not a single canned trajectory.

A.5 Annotated passage map for *The Black Sheep*

The translated story is not reproduced in full here because the text remains copyrighted. What can be included usefully is a passage map that identifies the parts of the story on which the model relies and the analytical object attached to each part. Figure 13 does that in paraphrased form.[2]

[2]The map is based on the sequence in Calvino (1993). It is intended as an interpretive guide to the model, not as a substitute for the story.

Story movement 1. Everyone steals at night and everyone is robbed in turn. Equality survives because outgoing predation and incoming predation offset one another.

Model fit. Section 3 turns this opening into the unit transfer benchmark and the balanced theft cycle of Definition 3.1 and Proposition 3.3.

Story movement 2. One man remains at home instead of stealing. The immediate effect is local, not universal. A planned burglar returns empty handed, and another household keeps what it otherwise would have lost.

Model fit. Lemma 3.4 records the exact first night accounting. Theorem 3.6 shows why that local windfall destroys stationarity once exposure rises with wealth.

Story movement 3. Inequality appears. Some households become rich because the broken chain no longer offsets gains and losses symmetrically.

Model fit. The proportional theft extension introduces wealth dependent exposure and turns class formation into a dynamic consequence, not a narrative assertion.

Story movement 4. The rich no longer steal personally. They hire the poor to steal for them.

Model fit. Lemma 3.7 and Proposition 3.8 derive the feasibility and surplus conditions for patronage. Figure 6 studies how that surplus moves with targeting advantage and outside options.

Story movement 5. Later the rich hire guards and police. Violence becomes more selective, but the unequal order remains.

Model fit. Theorem 3.10 gives the guarding threshold. The welfare analysis and Figures 7 through 11 show how organized protection can stabilize the spoils of the earlier transition.

Figure 13: Annotated passage map of the story by Calvino. The figure paraphrases the key narrative movements and indicates the point in the paper where each movement is formalized or simulated.